

Does job search assistance reduce unemployment? Evidence on displacement effects and mechanisms

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Abstract

Using a two-level randomized experiment we find that job search assistance (JSA) reduces unemployment among the treated, but also creates displacement effects. Analyses of mechanisms show that vacancy referrals from caseworkers to job seekers explain the positive effects for the treated by helping the job seekers apply to the most relevant jobs earlier. We also find that the overall assessment of JSA hinges on how the displacement effects hit the labor market and to what extent firms react by opening new vacancies. The displacement is larger in weak labor markets and we find no displacement of resources.

Keywords: vacancy referrals, counseling, job search, randomized experiment.
JEL codes: J68, J64, C93

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1 Introduction

Evidence suggests that job search assistance (JSA), aimed at helping job seekers search for jobs more efficiently, is one of the more powerful measures in the active labor market policy (ALMP) toolbox. As summarized in Card et al. (2010, 2018), both observational studies and randomized experiments, from a wide range of countries, document generally positive impacts of JSA. However, while JSA creates positive direct effects for the treated, it may also lead to substantial displacement of jobs for the non-treated. Although the concern for displacement effects has long been discussed in the literature, only a few recent studies have convincingly pinned down these effects empirically (Crépon et al., 2013; Gautier et al., 2018).¹ Since such displacement effects mainly represent a re-ordering of job queues they have important policy implications.

The fact that JSA is associated with positive effects for the treated and negative effects for the non-treated raises the question of how to design JSA programs with more favorable overall employment effects. In this paper we use a large-scale two-level randomized experiment (following Miguel and Kremer, 2004; Crépon et al., 2013) and unique administrative data to shed new light on this question. The experiment was conducted at local public employment offices in Sweden and introduced extra meetings with a caseworker during the first quarter of unemployment for all newly unemployed job seekers. The two-level randomization over both offices and job seekers identifies if the treated job seekers are more likely to find a job (direct effect) as well as to what extent this comes at the expense of non-treated job-seekers in the same labor market (displacement effect).

Initially, we document both positive direct effects of JSA for the treated and substantial negative displacement leading to higher unemployment for the non-treated. While the exit rate from unemployment for treated job seekers increases by 4.6 percent, it falls by 3.8 percent for the non-treated. Failure to account for displacement effects leads one to overstate the average impact of JSA on the treated by 89%. These findings are in line with the results in the recent displacement literature (Crépon et al., 2013; Gautier et al., 2018), and confirm that it is important to take displacement effects into account when evaluating labor market policies.

Our main contribution is that we use exceptionally rich administrative data on how the Public Employment Service (PES) helps the job seekers search for jobs and on job seekers' search behavior to dissect why the JSA program helps the treated job seekers and the types of displacement effects at work. We first examine why the JSA program was comparably effective for the treated and show that the positive direct effect was due to an increased number of vacancy referrals passed on from caseworkers to job seekers. Caseworkers use their expertise to find and point job seekers to relevant job openings. By tracking search behavior, we see that job seekers use and benefit from the vacancy information when searching for jobs. The treated job seekers do not broaden their search in terms of occupations or geographical distance. Instead, the vacancy referrals lead to a more streamlined job search process by helping job seekers apply to the most relevant jobs earlier. We also document that our JSA program, focusing on vacancy referrals, is relatively cheap compared to previously studied JSA interventions.

Technological advancements may make distance meetings an increasingly important element of

¹Earlier evidence on displacement includes Blundell et al. (2004), Pallais (2014), Ferracci et al. (2014), Lalive et al. (2015) as well as Dahlberg and Forslund (2005) and Albrecht et al. (2009) for Sweden. Despite these concerns most previous JSA studies focus on effects for the treated and neglect job displacement and other equilibrium effects (see, e.g., the overviews by Heckman et al., 1999; Kluve, 2010; Card et al., 2018).

future JSA programs, for instance, in case of long travel times or travel restrictions (e.g., during pandemics). We exploit that the offices in our experiment were randomly assigned to provide face-to-face, distance or group meetings, and find that face-to-face and distance meetings have equally large positive employment effects, supporting the case for communication technologies when providing JSA.

Another key contribution of our paper is that we provide evidence on the types of displacement effects at work. Initially, we examine resource allocation, and document that the negative effects for the non-treated do not arise because resources are allocated away from them. This is bad news for policy as resource allocation can be avoided by considering the funding arrangements, whereas displacement of jobs is more challenging to address. It also means that the non-treated do not receive fewer referrals, which implies that each vacancy gets more referrals, increasing the competition for the available jobs and creating the displacement we observe.

We also use local vacancy information and find suggestive evidence of a delayed vacancy response, with larger changes in the vacancy rate towards the end of the experiment. It suggests that JSA may have more positive effects in the long run, because in the short run the program mainly leads to search congestion, while in the long run improved search induces firms to open more vacancies, which increases the job finding rates and pushes down the unemployment rate. The implications of a full-scale roll-out of the JSA program and the delayed vacancy response are examined in a structural analysis, building upon the Diamond-Mortensen-Pissarides (DMP) model in Gautier et al. (2018). Our simulations show that increasing the treatment share from 0 to 100 percent lowers the unemployment rate by around 0.2 percentage points, suggesting that the net effect of a full-scale roll-out is positive despite the substantial displacement. Welfare is decreasing in the share of participants, mainly because the program implies lost non-market time. However, when we incorporate a delayed vacancy response into the structural analysis the welfare effects are reversed to positive, showing that the overall assessment of JSA hinges on firm side responses and vacancy creation. This is interesting given the large displacement documented by Crépon et al. (2013) and Gautier et al. (2018).

We finally aim to deepen our understanding of local labor market conditions and the impact of JSA. We document that the vacancy information helps the treated job seekers in both tight and slack labor markets, but the displacement is more extensive in weak labor markets compared to strong labor markets with no job rationing and good access to alternative jobs. We also explore whether the impact of JSA depends on the search intensity in the local labor market before the program.

Our paper relates to several strands of the literature. The only previous study using a two-level randomization approach to estimate displacement effects of JSA is the pioneering study by Crépon et al. (2013). They analyze a JSA program provided by private firms to young, long-term unemployed college graduates in France, and find that the positive effects for the treated were smaller than the negative displacement effects for the non-treated. The fact that our experiment targets all newly unemployed workers may explain why we find more positive overall effects of JSA. Another recent study of displacement is Gautier et al. (2018), who exploit a JSA intervention targeted at newly unemployed job seekers in two Danish regions using a difference-in-differences approach. The main contribution of our study in relation to these papers is that we deepen the understanding of important mechanisms. This includes the new evidence on the role of vacancy referrals and its impact on job-search behavior, the analysis of resource allocation, and the evidence on the delayed vacancy response and its implication for the long-run effects of JSA. We also confirm the finding in Crépon et al. (2013)

that displacement is stronger in weak labor markets.

Our paper also contributes to the existing experimental evidence on the direct effects of JSA for the treated. Many previous experiments examine interventions that mix different program components, which makes it hard to pinpoint exactly why the interventions help the unemployed. Examples include interventions with both early meetings and activation programs (e.g., Graversen and van Ours, 2008; Gautier et al., 2018), interventions with both support and monitoring (e.g., Dolton and O’Neill, 2002; van den Berg and van der Klauuw, 2006; Manoli et al., 2018), and interventions with JSA support during both the unemployment spell and upon entering employment (Crépon et al., 2013). In addition, most studies have been unable to study how JSA affects caseworker and job seeker behavior, and how this operates to the employment effects.² We show that vacancy referrals create the behavioral responses behind the positive direct effects for the treated.

The literature comparing different meeting formats when providing JSA is surprisingly limited. Although interventions differ across studies, few studies compare the effectiveness of different types of meetings. We are not aware of any previous studies on JSA using communication technologies. One randomized study that compares meeting formats is Maibom et al. (2017), but in their case each treatment was given in only one region, whereas we have a randomization design that explicitly allows for a comparison of different meeting formats.³

We proceed as follows. Section 2 details the experiment and Section 3 presents the data and the empirical strategy. Section 4 gives the main results. Section 5 examines why JSA help the unemployed, and Section 6 investigates the origin of the displacement effects. Section 7 concludes.

2 The experiment

2.1 The JSA program

Job search assistance aims at enhancing job search efficiency and includes measures such as job search courses, counseling, vocational guidance and monitoring. A straightforward way to intensify JSA is to have more meetings between job seekers and professional caseworkers (see, e.g., Graversen and van Ours, 2008). In line with this, the JSA program we study more than doubled the meeting frequency during the first quarter of unemployment. The extra meetings were of three types: individual face-to-face meetings, individual distance meetings and group meetings. The program was focused on job seekers and not on firms, and had no explicit aim to create vacancies.

Face-to-face and distance meetings were similar in that they added three extra meetings to the baseline services of about two meetings within the first quarter (a shorter registration meeting and a longer planning meeting) and that they focused on personalized job search assistance, without any detailed instructions to caseworkers. The distance meetings—online or via telephone—aimed at creating greater flexibility for caseworkers and job seekers. Group meetings used a more detailed protocol, gathering 10–15 job seekers in seminars on topics such as CV writing, interview training, and

²A recent exception is Arni (2015), who combines a randomized experiment with survey data to study behavioral mechanisms of an intensive counseling program targeted at older job seekers at two PES agencies in north-western Switzerland.

³Our paper also relates to some previous studies that examine other aspects of referrals. For instance, van den Berg et al. (2019) focuses on how vacancy referrals interact with monitoring and benefit sanctions, whereas the referrals in our study aim at helping job seekers to more quickly find the relevant jobs. There are also studies focusing on other types of referrals, such as firms relying on referrals from current workers (e.g., Heath, 2018).

network creation. Five seminars were to be held during the first two weeks and thereafter, participants were divided into smaller groups supposed to meet weekly for two months (not visible in data). Group meeting participants were informed about all five meetings immediately, while participants in face-to-face and distance meetings were typically summoned by caseworkers to one meeting at the time (on average 8 days before the meeting). All meetings were mandatory,⁴ but only job seekers with unemployment benefits could be sanctioned if not showing up when summoned.

To estimate the costs of our JSA program we sent out a detailed time-use survey to all involved caseworkers. Let us start with the face-to-face meetings. Caseworkers were asked to estimate the average time spent on a meeting, including time for preparation (25 minutes), time spent on the meeting (27 minutes), and time spent on documentation and other activities after the meeting (19 minutes). We also asked about the time lost due to cancelled meetings (2 minutes for each held meeting). Thus, caseworkers spent on average 73 minutes per meeting, amounting to 219 minutes for the three additional face-to-face meetings. This corresponds to 2.5% of the monthly working hours of a caseworker, who on average earned SEK 29,000 per month in 2015. Adding overhead costs for premises, administration and managers (estimated to roughly 30%), this gives an estimated cost of SEK 914, or 99 euros, per job seeker for the face-to-face meetings. Similar calculations give an estimated cost of SEK 960 (104 euros) for distance meetings and SEK 465 (50 euros) for group meetings (assuming 12.5 job seekers per meeting). The average cost across the three meeting formats is SEK 780 (85 euros).

Although the JSA program more than doubled the meeting frequency, it was less intense and less costly than many previous JSA programs (see, e.g., the studies surveyed in Card et al., 2018)—in particular the programs evaluated in recent displacement studies. The JSA program in Crépon et al. (2013) was operated by private providers. Caseworkers met with the job seekers at least once a week during the first six months of unemployment and thereafter gave continued support during six months. The maximum total payment ranged from 1,600 to 2,100 euros per job seeker—up to 20 times the cost of our program. The JSA program in Gautier et al. (2018) and Graversen and van Ours (2008) consisted of a job-search program, meetings with a caseworker and activation programs, which costed about 285 euros per job seeker, which is more than 3 times the costs of our JSA program.

2.2 Randomization and program implementation

The experiment took place during six months in 2015 (March–May and August–November), with a halt during the summer. It involved 72 Public Employment Service (PES) offices—one fourth of all offices in Sweden. The offices were compensated by central project-specific funding, with the intention to fully finance the program, and in Section 6.1 we see no evidence of any crowding-out of resources at the local offices, suggesting that this goal was achieved.

The offices were selected to constitute a representative sample in terms of geographical dispersion and size. Offices with a monthly inflow of less than 20 job seekers were excluded, since they were too small for group meetings. We divided offices into metropolitan areas, medium-sized cities and rural areas, and created six-folds with similar economic and demographic conditions. Since many local labor markets in Sweden have one employment office, randomization over offices in many cases implies randomization over local labor markets, facilitating estimation of displacement effects.⁵ The

⁴The exception was distance meetings during spring, when telephone meetings could not be used due to security issues. Job seekers without access to a computer with internet connection and an e-ID could therefore not participate.

⁵We see that 70% of job seekers find jobs in the local PES office catchment area, showing that many indeed find

main exception is metropolitan areas with one large labor market but several offices. To address this, we selected metropolitan-offices where job seekers were less likely to compete for the same jobs, e.g., in different parts of the city. In total we have 12 six-folds (72 offices): 3 six-folds from metropolitan areas, 6 six-folds from medium-sized cities and 3 six-folds from rural areas.

The target population consisted of all newly unemployed job seekers, only exempting job seekers who had been unemployed in the last three months and newly arrived immigrants. We applied a two-level randomization strategy over both offices and job seekers. Randomization over job seekers within active offices identifies the direct effect of the JSA program by comparing treated and non-treated job seekers in the same labor market, and randomization over offices identifies the displacement effects by comparing non-treated job seekers at active and non-active offices. We randomized offices within each strata (six-fold) of similar offices and assigned one office face-to-face meetings, one distance meetings, one group meetings, and the remaining three to be non-active offices—in total 36 active offices (12 per meeting format) and 36 non-active offices that continued with the baseline services.

The active offices randomly assigned job seekers to the treatment group, based on the date of birth. The treatment share was randomly set to 50 or 80% during spring, but we had insufficient power to exploit this difference. During fall, the treatment share was 50% in all active offices. The overall treatment share, calculated from the last row in Table 1, was approximately 53%. We use the random assignment to perform balancing tests in Table 1, accounting for differences in treatment shares across offices.⁶ Columns 1 and 2 present averages for treated and non-treated job seekers at the active offices and column 3 for job seekers at the non-active offices. Columns 4 and 5 show p-values for comparisons between the groups. The results are reassuring: there are no significant differences between the treated and the non-treated job seekers at the active offices, nor between job seekers at active and non-active offices.

Table 2 looks at the program implementation, by showing the intention-to-treat comparison of treated and non-treated job seekers (according to the treatment protocol by date of birth) at the active offices, for the full sample and the three meeting formats. Panel A shows information about program participation collected from the project-specific database and written acts at the PES. During the first registration meeting, caseworkers were instructed to use an experiment-specific tool that revealed the treatment status of the job seeker. Job seekers assigned to treatment were then informed about the program by the caseworker, whereas no information was given to the non-treated. Defining a variable that takes the value one if the experiment-specific tool was used and the revealed treatment status was treated, else zero, the estimate in the first line of panel A implies that 62% of the treated job seekers were informed about the program. Attrition is mainly explained by individual caseworkers failing to use the tool, and some offices failing to correctly capture the target population. However, since we observe both the entire target population and the sub-population informed about the program, we have full control over how the experiment was implemented. Panel A further shows that 23% of the target population participated in at least one program specific meeting;—37% of those informed about

jobs locally. Regarding data details, the catchment areas do not map with the municipality of residence, which is the finest level of information on residency in our data. We therefore define the office catchment area by the municipalities in which more than 10 percent of the target population during 2012–2015 registered at that office.

⁶To correct for differences in treatment shares across offices (e.g., 50% and 80% during the spring), we weight each treated individual by the inverse of the intention to treat share, $1/w$, where w is the observed share of job seekers at the local office who would be randomized to treatment based on their date of birth. Analogously, we weight each non-treated job seeker by $1/(1 - w)$. This gives treated and non-treated job seekers at each office the same weight in all comparisons, which corrects for differential composition of job seekers across offices. The weighting is used throughout the paper.

the program (0.23/0.619)—and that the share was highest for face-to-face meetings (27%), and lowest for group meetings (16%). Reasons for not initiating treatment was, for example, caseworkers failing to offer the extra meetings, job seekers finding a job before the first meeting, or job seekers simply not showing up for the extra meetings.

Table 1: Summary statistics by treatment status

Variables	Active offices		Non-active offices	p-values of balancing tests	
	Treated job seekers	Non-treated job seekers		Treated vs Non-treated job seekers at Active offices	Job seekers at Active vs Non-active offices
	(1)	(2)	(3)	(4)	(5)
Age	33.33	33.39	33.54	0.707	0.637
Male	0.542	0.539	0.555	0.571	0.186
Unemployment benefits take-up	0.642	0.638	0.634	0.546	0.747
Functional disability	0.052	0.052	0.054	0.931	0.602
Matchable	0.868	0.862	0.864	0.117	0.936
Education level					
Less than high school	0.224	0.223	0.225	0.873	0.878
High school	0.491	0.493	0.486	0.833	0.690
College	0.285	0.285	0.289	0.932	0.844
Region of birth					
Sweden	0.678	0.667	0.641	0.059	0.399
Nordic countries	0.013	0.015	0.015	0.197	0.547
West Europe	0.036	0.034	0.036	0.256	0.857
Outside west Europe	0.273	0.284	0.308	0.035	0.385
Unemployment days					
Year $t-1$	30.66	30.43	30.98	0.767	0.777
Year $t-2$	67.42	68.49	68.25	0.449	0.914
Year $t-3$	69.57	71.83	71.22	0.133	0.868
Year $t-4$	63.82	63.68	66.14	0.921	0.438
Number of unemployment spells					
Year $t-1$	0.431	0.440	0.442	0.427	0.793
Year $t-2$	0.789	0.800	0.793	0.518	0.972
Year $t-3$	0.806	0.815	0.819	0.630	0.840
Year $t-4$	0.706	0.721	0.716	0.397	0.960
Labor market programs, last 4 yrs					
Labor market education	0.024	0.020	0.022	0.172	0.965
Preparatory education	0.048	0.047	0.047	0.869	0.983
Labor market training	0.027	0.030	0.031	0.389	0.370
Subsidized employment	0.106	0.108	0.106	0.711	0.920
Observations	14,075	12,463	31,240	26,538	57,778

Notes: Summary statistics by treatment status, weighted by the treatment share within each office (see footnote 6). The sample includes the active and non-active PES offices during the experiment period. *Matchable* indicates the caseworker's initial assessment about the job seeker's potential to take a job on short notice. *Unemployment days* and *Number of unemployment spells* are presented for the four calendar years $t-1$ to $t-4$ before the start of the current unemployment spell in year t . Standard errors in column 5 are clustered at the PES office level.

Panels B and C of Table 2 show that the impact of the program on the frequency, type and time pattern of meetings was consistent with the treatment protocol. Information about meetings was collected from the PES administrative registers; *physical meetings* include face-to-face meetings and group meetings, *distance meetings* include contacts over telephone and online using an e-ID, and *all meetings* are the sum of these. Panel B shows that the treated job seekers received on average 0.5 more meetings, which, given the take-up rate of 23%, corresponds to about two more meetings for those who actually showed up (not adjusted for early exits). The meeting frequency increased during the experiment (quarter 1) but not after the experiment (quarter 2), and the increase was similar

for face-to-face and distance meetings and slightly larger for group meetings. Panel C shows that job seekers assigned to face-to-face meetings and group meetings received more physical meetings, whereas job seekers assigned to distance meetings received more distance meetings.

Table 2: ITT effects of the JSA program on program participation and meetings in active offices

	All meeting types (1)	Face-to-face meetings (2)	Distance meetings (3)	Group meetings (4)
Panel A: Program participation				
Informed about the program	0.619*** (0.004)	0.601*** (0.007)	0.621*** (0.008)	0.643*** (0.008)
At least one program meeting	0.230*** (0.004)	0.265*** (0.006)	0.247*** (0.007)	0.164*** (0.006)
Panel B: Number of meetings				
All meetings quarter 1	0.500*** (0.026) [3.159]	0.414*** (0.039) [3.150]	0.418*** (0.046) [3.302]	0.707*** (0.052) [3.019]
All meetings quarter 2	-0.017 (0.021) [1.176]	0.002 (0.033) [1.152]	-0.034 (0.039) [1.247]	-0.024 (0.039) [1.131]
Panel C: Type of meeting				
Physical meetings quarter 1	0.318*** (0.021) [2.262]	0.376*** (0.032) [2.287]	-0.015 (0.035) [2.331]	0.596*** (0.043) [2.155]
Distance meetings quarter 1	0.182*** (0.015) [0.897]	0.038* (0.022) [0.863]	0.433*** (0.030) [0.971]	0.112*** (0.027) [0.864]
Observations	26,538	10,567	8,259	7,712

Notes: Regression of each variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes the active PES offices during the experiment period. *Informed about the program* indicates that the experiment-specific randomization tool was used and that the tool revealed the treatment status to be treated. Information about *At least one program meeting* collected from written acts at the PES. *Physical meetings* include visits, *Distance meetings* include contacts over telephone or online, and *All meetings* are the sum of the two. Standard errors in parentheses, control means in square brackets. *** p<0.01, ** p<0.05, * p<0.1.

3 Data and empirical strategy

3.1 Data

Our analysis benefits from access to rich data. We provide an overview of the data in this section and give more details in the analyses. From the Swedish PES we have information on individual characteristics, all meetings between caseworkers and job seekers and daily records of unemployment status (including participation in active labor market programs) for all job seekers. We also use data from a project-specific database that was set up for the experiment, and information extracted from written acts at the PES. We analyze the impact of the JSA program on the exit rate from unemployment within the first quarter, as well as the number of days in unemployment (including re-unemployment) during the first quarter and the first year.

Detailed information about caseworkers' actions and job seekers' search behavior allows us to study mechanisms behind our results. For caseworker actions, we use several administrative data sources

from the PES, including vacancy referrals from caseworkers to job seekers, registered violations of the job search rules and information from individual action plans. For job seeker search behavior, we are the first to use data from monthly activity reports. Since September 2013, job seekers in Sweden are required to submit a monthly report on the number and type of all job search activities to the PES (e.g., job applications and search channels). Failure to provide a report, or submitting a report indicating too low level of search activity, may lead to a benefit sanction. Moreover, for each job application, the job seekers have to report the name and telephone number of the firm, and the occupation to which they apply, so that caseworkers can validate the information in the reports. Therefore, the activity report data should be a reliable data source for measuring job search behavior.

We use employment records and wage data from Statistics Sweden. We also use data on all vacancies posted at the PES per municipality and month over the period 2012–2016.⁷

3.2 Empirical strategy

We first analyze the data the way it is usually done in evaluations of experiments: we utilize the randomization over job seekers within offices and compare the treated and non-treated job seekers at the active offices. We assign treatment status according to the treatment protocol, i.e., by the job seekers' date-of-birth, and estimate:

$$Y_i = \alpha_0 + \beta_0 1(\text{Assigned to program}_i) + \rho X_i + \varepsilon_i, \quad (1)$$

where the parameter of interest is the intention-to-treat (ITT) effect of the JSA program, β_0 . Since individuals are randomized within offices this comparison of treated and non-treated job seekers at the active offices gives the direct effects of the program, and we do not need to cluster at the office level. As a robustness check we also present results controlling for job seeker characteristics, X .⁸

We then estimate displacement effects by exploiting the randomization over offices. Here, the model for individual i in office j is:

$$Y_{ij} = \alpha_1 + \beta_1 1(\text{Assigned to program}_{ij}) + \beta_2 1(\text{In a program area}_j) + \rho X_i + \varepsilon_{ij}, \quad (2)$$

where *In a program area* indicates an active office and *Assigned to program* indicates being in the treatment group. In this analysis, exploiting the randomization across offices, standard errors are clustered at the PES office level. The displacement effect is given by β_2 , which is identified through the comparison of non-treated job seekers at active and non-active offices. β_1 captures the direct comparison of the treated and the non-treated at the active offices, and the net effect for the treated job seekers, compared to non-treated job seekers at the non-active offices, is given by $\Delta \equiv \beta_1 + \beta_2$. The overall effect is the average across treated and non-treated job seekers at the active offices compared to job seekers at the non-active offices. In Section 4, we also use pre-experiment data and an extended model to increase the precision of our displacement effect estimates.

We will also analyze mechanisms in various ways. This includes why JSA helps the unemployed, a comparison of meeting formats, effects for different groups, displacement of resources, displacement

⁷All vacancies are not posted at the PES, but note that firms looking to recruit from the pool of unemployed tend to post at the PES, suggesting that vacancies that are relevant for the job seekers in our experiment are posted at the PES.

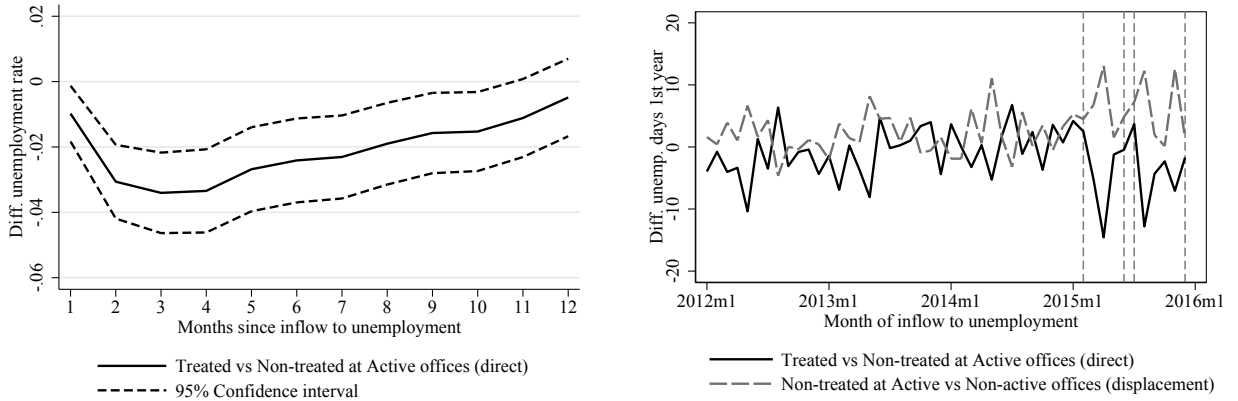
⁸We focus on the ITT effect of the JSA program, but in Appendix B we also report instrumental variables (IV) estimates of the LATE of program participation.

in the short and the long run, and displacement in different types of labor markets.

4 Effects of the job search assistance program

A visual inspection of the direct ITT effect of the JSA program in Figure 1(a) shows a clear difference in unemployment rates between treated (assigned to the JSA program) and non-treated (assigned to the control group) at the active offices. The effect reaches its maximum of about three percentage points when treatment ends, after 3–4 months. The immediate effect is expected since the program aims at helping job seekers find jobs faster, precluding any lock-in effects. Although the effect decreases after treatment ends, it is still significant until the tenth month since inflow to unemployment. The visual evidence is confirmed by the regression estimates in columns 1–2 of Table 3, reporting the direct ITT effect, β_0 , from equation (1). It shows that the JSA program increases the probability of leaving unemployment during the first quarter of unemployment by 3.5 percentage points, or about 10% (column 2 with individual characteristics).⁹ From Table A-1 in Appendix A we also see that the treated have 1.5 fewer days of unemployment during the first quarter, and 5.9 fewer days during the first year.

Figure 1: Difference in unemployment between treatment groups



(a) Difference in unemployment rates between treated and non-treated job seekers at active offices

(b) Difference in unemployment days during 1st year of unemployment between treated and non-treated job seekers at active offices and between non-treated job seekers at active and non-active offices

Notes: All new job seekers per office and month, during the experiment in 2015 in Panel A and during 2012–2015 in panel B. Treatment status defined by date of birth, in Panel B it is the status the job seeker would have had if the program had taken place at that time. An active office is an office that was randomized to implement the JSA program in 2015. A non-active office is an office randomized to be non-active during the experiment in 2015. The dashed lines in Panel A indicates 95% confidence intervals with robust standard errors. The dashed vertical lines in panel B indicate the two experiment periods.

We next take displacement effects into account. The program area estimate from equation (2) in column 3 of Table 3 indicates a sizeable displacement effect, but the precision is low. To increase

⁹This exploits daily data on exits from unemployment. We can connect some of the data to employment records, albeit with a lower quality of time indicators than the unemployment data, and see that 88% of the exits have a registered employment spell within 3 months. Estimating the model on exits from unemployment within the first quarter, conditional on having a registered employment spell, gives a practically unchanged ITT effect, suggesting that the treated job seekers do not drop out of the labor market due to the extra meetings.

Table 3: Direct and displacement effects (ITT) of the JSA program on the exit from unemployment during the first quarter

	Experiment period in 2015			2012–2015	
	(1)	(2)	(3)	(4)	(5)
Assigned to program	0.036*** (0.006)	0.035*** (0.006)	0.035*** (0.005)	0.035*** (0.006)	0.034*** (0.006)
In a program area			−0.015 (0.010)	−0.016** (0.007)	−0.015** (0.006)
Net effect treated			0.020** (0.009)	0.019*** (0.006)	0.018*** (0.005)
Control mean	0.354	0.354	0.368	0.390	0.390
Year dummies	No	No	No	Yes	Yes
Month dummies	No	No	No	Yes	Yes
PES office dummies	No	No	No	Yes	Yes
Controls	No	Yes	Yes	No	Yes
Clusters	No	No	72	72	72
Observations	26,538	26,538	57,778	552,816	552,816

Notes: Regression of each outcome variable on an indicator for active PES office (“In a program area”) and an indicator for active PES office $\times$ intention to treat status is treated (“Assigned to program”), weighted by the treatment share within each office (see footnote 6). Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, clustered at the PES office level in columns 3–6. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

precision, we add data from time periods prior to the experiment.¹⁰ The strategy is illustrated for the number of unemployment days during the first year in Figure 1(b), with the solid black line comparing treated and non-treated job seekers at active offices and the dashed grey line comparing non-treated job seekers at active and non-active offices, by calendar month of inflow to unemployment. For the experiment period these are the actual treated and non-treated, and before the experiment the “treated” are those who would have been treated if the treatment protocol with randomization based on date of birth had been applied to the inflow during that period. The dashed vertical lines indicate the two experiment periods. Before the experiment there is no systematic pattern in the data—the time series are noisy but stay around zero. During the experiment periods, unemployment decreases for the treated relative to the non-treated at the active offices, and increases for the non-treated at the active offices relative to the non-treated at the non-active offices. The latter gives a clear indication that the JSA program caused displacement effects.

We next use a regression model and the pre-experiment data to examine these displacement effect patterns. For individual i in office j starting unemployment in year y and month m we estimate:

$$Y_{ijym} = \delta_j + \delta_y + \delta_m + \beta_1 1(\text{Assigned to program}_{ijym}) + \beta_2 1(\text{In a program area}_{ijym}) + \rho X_i + \varepsilon_{ijym}, \quad (3)$$

where the variables *Assigned to program* and *In a program area* are set to zero for all job seekers in the pre-period, δ_j are office fixed effects, and δ_y and δ_m are year and month dummies. Standard errors are clustered at the employment office level. This strategy of using pre-experiment data to capture office level heterogeneity assumes similar employment trends at the treated and non-treated offices in absence of the experiment. Figure 1(b) reveals no diverging trends before the experiment, lending

¹⁰We include job seekers during 2012–2015 using the same sample restrictions for the target population as during the experiment in 2015. We begin in 2012 since several reforms were introduced between 2011 and 2012, including public investments to increase the number of caseworkers, and a new mandate for the PES to provide services to immigrants.

some support to the assumption.

The estimates of model (3) in columns 4–5 of Table 3, using pre-experiment data, are similar to using only data from the experiment period in column 3, but due to the increased precision the standard errors are smaller, and we can conclude that there are displacement effects. Our preferred estimates in column 5 (with individual characteristics) suggest that the program reduces the exit rate for the non-treated job seekers at the active offices by 1.5 percentage points (3.8%) during the first quarter of unemployment, a finding that is consistent with substantial displacement effects of the JSA program. The direct ITT effect for treated compared to non-treated at active offices is 3.4 percentage points, but when taking the displacement effect for non-treated into account, the net ITT effect for treated job seekers compared to non-treated job seekers in non-active offices, $\beta_1 + \beta_2$, is only 1.8 percentage points, or 4.6%. A regular randomized controlled treatment (RCT) evaluation, not taking displacement into account, would thus overestimate the average ITT effect by almost 89% (with a standard error of 15.4%, using the delta method).

Accounting for both the positive effect on the treated and the negative displacement effect on the non-treated, in combination with the treatment share of 53%, we can calculate the overall effect across treated and non-treated job seekers at the active offices. The overall effect is positive but small, implying that the JSA program increases the job finding rate by 0.25 percentage points or 0.7% on average across all job seekers at the active offices.^{11,12} A regular RCT evaluation, ignoring displacement effects, would assume an overall effect of 1.8 percentage points (the direct effect for the treated is 3.4 percentage points and 53% are treated), which would grossly overstate the true overall impact. Given the costs of the program, ignoring the displacement effects may therefore have led to incorrect policy conclusions.

Crépon et al. (2013) find a net effect for the treated close to zero, insignificant, and smaller than the displacement effects for the non-treated, implying that more jobs were lost than found. Hence, we find somewhat more positive overall effects of JSA. One reason could be that we study a more general target group and a public JSA program, whereas they study young college graduates unemployed for at least six months and private provision of JSA. Although our treatment substantially boosted employment for program participants, we find smaller direct effects of JSA than Crépon et al. (2013)—most likely because we study a less intensive intervention. Graversen and van Ours (2008) find that the job finding rate in the treatment group is 30% higher than in the control group, which is larger than the direct effect of our program, but the Danish program was also more intense (see the discussion in Section 2.1). Gautier et al. (2018), similarly to our study, document extensive displacement effects of JSA from the same Danish experiment. Schiprowski (2020) finds that job seekers who lose a meeting with their caseworker stay unemployed about 5% longer, which is of similar magnitude as the IV

¹¹The overall effect is 0.53×1.8 percentage points (effect for the treated) + 0.47×-1.5 percentage points (effect for the non-treated) = 0.25 percentage points.

¹²Our design allows to study the overall effect of the program for the target population, but this ignores displacement for non-treated job seekers *outside* the target population. Table A-2 in Appendix A presents displacement effect estimates for three groups outside the target population: the stock of job seekers that entered unemployment before the experiment, the inflow of unemployed during the summer months when no one was assigned to the program, and the inflow of unemployed during the experiment period that did not belong to the target population. The results indicate that these groups were unaffected, most likely because they do not compete for the same jobs as the target population. Also job seekers at other PES offices could suffer from displacement, if they search for jobs in the same labor market as the treated job seekers. The research design does not allow us to study these effects, but since 70% of job seekers achieve employment in the PES office catchment area (see footnote 5) and the breadth of search was unaffected (see Table 5), we think this is of minor importance.

estimates that we report in Appendix B.

5 Why does JSA help the unemployed?

To shed light on why the direct effect of the JSA program for the treated arises, we exploit the richness and the quality of the data and study both actions taken by the caseworkers and accompanying behavioral reactions by the job seekers. Data on caseworker actions are available from administrative records at the PES for the entire target population. Data on job seeker behavior from monthly activity reports filed by job seekers are only available for those who are still unemployed when the report is supposed to be submitted, who are supposed to report according to PES registers, and who actually submitted their report.¹³ Panel B of Table 4 shows that there is no difference between the treated and the non-treated in the propensity to report. Since we are interested in understanding the direct effect of the JSA program, this section only uses data on the treated and non-treated job seekers at the active offices during the experiment. For each analysis, we use the model in equation (1), and study the first quarter of unemployment, during which the JSA program took place.

5.1 Vacancy information and job search

Vacancy referrals from caseworkers to job seekers may speed up the exit from unemployment by broadening job search or by pinpointing job seekers to the most relevant job openings. The results in Panel A of Table 4 suggest that vacancy information is a key component behind the positive direct effects of the JSA program. Columns 1–3 show that the JSA program raised the total number of vacancy referrals passed on from caseworkers to job seekers by 0.10 or 5% during the first quarter, both due to more mandatory referrals (up 16%) and more suggested job openings (up 4%). Columns 4–6 show that the job seekers took advantage of the vacancy information provided by the caseworkers, with the number of applications to referred vacancies increasing by 0.10 (23%) during the first quarter. The fact that both sources of information—administrative records of caseworker actions and the monthly reports on job seekers’ search behavior—paint the same picture gives strong support for the importance of the vacancy referral channel.

We take this as evidence that caseworkers use their expertise to prepare suitable vacancies, point the job seekers to these vacancies during the extra meetings and that the job seekers follow the advice and apply to the jobs, and this explains the positive direct effects of the JSA program. We can also relate the increase in the number of referrals provided by the caseworkers (5%) and referrals applied to (23%) to the increase in the number of meetings (16%, see Table 2). One interpretation of relative differences between these numbers is that caseworkers use the extra meetings within the JSA program to prepare highly suitable vacancies rather than focusing on giving as many vacancy referrals as possible, and that the job seekers see the value of these new vacancies and therefore apply to them to a higher extent than other referrals.

One potential concern is that the job seekers in the JSA program may report having used the referrals because they feel more closely monitored. However, several things speak against this interpretation. First, job seekers with unemployment benefits risk a sanction if they do not apply to

¹³We include everyone who submitted at least one report during the first quarter of unemployment, and take the average across the three possible reports during this period. Since some job seekers do not submit all three reports we include three indicators for whether the job seeker submitted a specific report or not.

mandatory referrals, but there is no such threat for suggested job openings.¹⁴ That both types of referrals increase therefore suggests that the job seekers did apply to the referrals and not just reported doing so (even if they did not) because of the fear of sanctions. Second, below we see no impact on the number of sanctions related to job search. It indicates that the caseworkers did not use the extra meetings to more closely monitor job search.

Table 4: Effects of the JSA program on caseworker actions and job seeker search behavior

Panel A:	Caseworker action: Vacancy referrals			Search behavior: Applications to vacancy referrals		
	Total (1)	Mandatory (2)	Suggested (3)	Total (4)	Mandatory (5)	Suggested (6)
Assigned to program	0.102** (0.046) [2.048]	0.029* (0.017) [0.181]	0.074* (0.041) [1.867]	0.104*** (0.027) [0.460]	0.029*** (0.008) [0.033]	0.075*** (0.026) [0.427]
Observations	26,538	26,538	26,538	11,959	11,959	11,959
Panel B:	Caseworker action: Monitoring			Search behavior: Search effort		
	Total notifications (1)	Notifications absence meeting (2)	Notifications search effort (3)	Prob. reporting (4)	Total activities (5)	Total job applications (6)
Assigned to program	0.025*** (0.008) [0.199]	0.019*** (0.003) [0.039]	0.001 (0.001) [0.008]	−0.006 (0.006) [0.520]	0.102 (0.337) [14.5]	0.020 (0.253) [7.81]
Observations	26,538	26,538	26,538	19,674	11,959	11,959
Panel C:	Caseworker action: Job search support and training			Search behavior: Search channels		
	Has action plan (1)	Support categories in plan (2)	Job search support categories (3)	ALMP participant (4)	Unsolicited job applications (5)	Other job enhancing activities (6)
Assigned to program	−0.006 (0.004) [0.886]	−0.009 (0.010) [1.445]	0.003 (0.007) [0.243]	0.004 (0.003) [0.066]	−0.113 (0.112) [2.546]	0.208 (0.145) [3.417]
Observations	26,538	23,431	23,431	26,538	11,959	11,959

Notes: Regression of each outcome variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes the active PES offices during the experiment period. All newly The outcome variables for caseworker action is the sum during the 1st quarter of unemployment, for search behavior the sum over the monthly activity reports filed in the first quarter. Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, control means in square brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table 5 next explores whether vacancy referrals affect unemployment duration by broadening job search in terms of occupation or geographical search area (see, e.g., Belot et al. (2019) for evidence on breadth of search), or by pinpointing the most relevant job openings, without changing the breadth of search. Columns 1–4 show the effect of the JSA program on the share of referrals within the job seeker’s preferred occupation (2/3-digits) and within the job seeker’s geographical area (region/office catchment area) during the first quarter of unemployment. The results show that breadth of search is unaffected by the treatment, which supports the second explanation above; that vacancy information—while leaving the direction of search unaffected—streamlines the search process by pointing job seekers

¹⁴The mandatory referrals are mainly used when caseworkers believe that there is a good match between the vacancy requirements and the job seeker’s skills.

to the most relevant jobs earlier. This also holds for occupation at the 1-digit level and for municipality of residence (not reported).

Table 5: Impact on breadth of search

	Share referrals within 2-digit occupation (1)	Share referrals within 3-digit occupation (2)	Share referrals within region (3)	Share referrals within catchment area (4)	Number of other referrals (5)
Assigned to program	0.000 (0.008)	0.001 (0.008)	-0.002 (0.006)	-0.001 (0.007)	-1.097** (0.448)
Control mean	0.537	0.493	0.830	0.580	10.44
Observations	13,234	13,234	13,234	13,234	13,234

Notes: Regression of each outcome variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes individuals with at least one vacancy referral at the active PES offices during the experiment period. Region is county of residence, catchment area is the area of the local PES office as defined in Section 2.2. Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In column 5 of Table 5 we study if caseworkers refer to vacancies that receive fewer referrals. For each job seeker and referral, we count the number of other referrals to the vacancy, and take the average per job seeker during the first quarter of unemployment. Caseworkers can observe the number of referrals per vacancy in the available IT systems at the PES, and could hence act on that information. Interestingly, the treated are referred to vacancies that receive about 1.1 or 10% less referrals, suggesting that the JSA program allows caseworkers to refer their job seekers to more relevant job openings with less competition from other referred job seekers. We found no evidence of caseworkers referring the treated job seekers to younger vacancies (posted more recently; not reported).

We next examine if the referrals affect job quality. Professional support from caseworkers with expert knowledge on the local labor market may help job seekers find higher-quality jobs, but the additional meetings may also pressure job seekers to accept jobs of lower quality or worse matches with employers. Table A-3 in Appendix A shows the effect on the log wage rate for the first job after unemployment (column 1) and on job duration (column 2), using data from Statistics Sweden on employment spells and on the full-time equivalent monthly wage rate.¹⁵ The estimates are insignificant and close to zero, supporting neither of these hypotheses. These findings suggest that caseworkers use vacancy referrals to get job seekers back to work faster rather than as a tool to improve job quality.

Section 2 showed that our JSA program was a relatively inexpensive intervention, suggesting that JSA programs with a strong focus on vacancy referrals is a cheap way to improve the job search process. Since our experiment did not include an explicit comparison of different types of JSA, we cannot conclude that JSA with a focus on vacancy information is the most efficient type of JSA. However, the low costs and positive employment effects indicate that vacancy information may help to improve other types of JSA interventions (e.g., job search courses, search clubs, and counselling).

¹⁵We estimate the effect on the re-unemployment rate within 5 months of exit from unemployment. The 5-month limit is due to the limited follow-up period in our data. The wage data cover roughly 50% of all jobs, including the entire public sector, large private firms and a sub-sample of small private firms. There is no difference across treated and non-treated with respect to the sampling scheme, conditional on finding a job.

5.2 Alternative interpretations

We now study other potential channels for the direct effect. Even if the aim of the program was to provide job search assistance, more meetings may increase monitoring. To study this we exploit that caseworkers should notify the unemployment insurance funds if they observe a violation of the rules.¹⁶ Column 1 of Panel B in Table 4 shows that the total number of notifications is higher for the treated job seekers, but column 2 shows that this is explained by more notifications due to absence from meetings, which was expected since the JSA program implies more meetings. In contrast, notifications regarding insufficient job search effort (column 3) is unaffected.

Although caseworkers did not use the extra meetings to intensify the monitoring of search effort, the increased notifications due to missed meetings may affect job seekers. However, using data on the sanction process, we find that only 28.8% of the notifications lead to a sanction and that the JSA program increased the probability of being sanctioned by only 0.1 percentage points. A reason behind the low sanction rate could be that we study newly unemployed individuals and the UI funds may be reluctant to issue sanctions for minor violations early in the unemployment spell. Note also that the sanctions are mild: the first sanction is a warning, and then 1, 5, and 10 days suspension of the benefits for the second to the fourth sanction. Since we study newly unemployed individuals, 79.8% of the sanctions only implied a warning. Even if we cannot rule out increased monitoring, the low sanction rate, the mild sanctions and the fact that the JSA program was not designed to increase monitoring suggest that it is not the main channel behind the effects we observe.

We have also examined the corresponding changes to job seekers' search behavior, and find no differences in terms of the total number of job-search activities (column 5 in Panel B of Table 4), or the total number of job applications (columns 6), showing that the notifications due to missed meetings did not increase search effort.¹⁷ Also, note that pre-program responses hardly explain the observed effects, since the difference in Figure 1 is relatively little during the first month and the median number of days until the first program meeting is 29 days.

Next, in Panel C of Table 4 we rule out that the effect stems from increased job search support and training. Caseworkers could, for instance, use the meetings to help job seekers write better CVs and job applications, prepare them for interviews, or provide valuable information about job search strategies. To study this, we use information from individual action plans, which specifies both the type of support and training that the PES commits to offer, and what actions each job seeker should undertake to find a job. The plan consists of up to eight different support categories.¹⁸ According to PES guidelines, an action plan should be set up in the beginning of the unemployment spell and revised as the job search continues. We can only study these outcomes for job seekers with an action plan. Column 1 of Panel C in Table 4 shows that 89% of the job seekers have an action plan, and there is no difference between treated and non-treated in the likelihood to have an action plan (column 1),

¹⁶The funds then decide about benefit sanctions, which are monetary fines (suspension of unemployment benefits). Refusal of job offers, insufficient job search, not showing up for meetings with PES, not applying for assigned jobs, and job quits may lead to a sanction. Similar rules apply to job seekers with other types of benefits.

¹⁷DellaVigna et al. (2020) show that search intensity tends to drop after job acceptance. Since the treated find jobs faster than the non-treated, this may confound the comparison of job intensity between the two groups. To test for this, we re-analyzed search intensity by excluding the month immediately prior to exit from unemployment, but this gave similar results (not reported).

¹⁸The categories are: Search for jobs, Improve your search, Guidance to work, Education to work, Start new business, Clarify your qualifications for work, Adapt your work situation, and Work preparatory measures. We define Improve your search and Guidance to work to be categories that specifically capture job search training.

the total number of support categories in the plan (column 2), or the number of support categories related to job search training (column 3). We also find no differences regarding the probability of participating in an active labor market program (column 4), and columns 5–6 show little impact on job seekers’ search channels (unsolicited job applications and other job-enhancing activities). All this evidence speaks against job search support and training as important channels.

5.3 Results across meeting formats and job seekers

The randomization of meeting formats across offices allows us to make credible comparisons of face-to-face, distance and group meetings. We have confirmed that the groups are balanced for each meeting type (not reported). The results on unemployment outcomes in Table 6 are not significantly different across formats, but the coefficients are larger for face-to-face and distance meetings than for group meetings. However, since the cost calculations in Section 2.1 showed that group meetings were less expensive we cannot conclude that group meetings are less effective.¹⁹ Interestingly, the estimates for face-to-face and distance meetings are of similar magnitude, which is encouraging in terms of the opportunities for communication technologies. While recent advancements will likely make distance meetings an increasingly important element of future JSA programs, we are not aware of any previous studies on this topic.

Table 6: Effects of the JSA program on unemployment, by meeting format

	Face-to-face meetings (1)	Distance meetings (2)	Group meetings (3)
Exit unemployment 1st quarter	0.043*** (0.010) [0.329]	0.034*** (0.011) [0.367]	0.024** (0.011) [0.375]
Unemployment days 1st quarter	−2.018*** (0.531) [75.92]	−1.397** (0.583) [73.40]	−0.952 (0.611) [73.30]
Unemployment days 1st year	−6.894*** (2.440) [204.4]	−7.307*** (2.609) [191.9]	−3.463 (2.766) [189.3]
Observations	10,567	8,259	7,712

Notes: Regression of each outcome variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes the active PES offices during the experiment period. Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, control means in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The existence of displacement effects in our experiment raises the important question of *who* gains and *who* loses. From a welfare perspective we may accept displacement if job search assistance benefits job seekers who are less attached to the labor market at the expense of those who are closer to finding a job. Given that a key component behind the direct effect is vacancy referrals from caseworkers to job seekers, we may expect a larger impact for disadvantaged job seekers who are less informed about how to search for jobs or have smaller social networks. Earlier evidence also suggests that JSA is particularly beneficial for disadvantaged workers (see, e.g., Card et al., 2018). Panel A of Table A-4 in Appendix A shows a significantly larger reduction of the number of days in unemployment in the first

¹⁹In related studies Maibom et al. (2017) find smaller employment effects for group meetings, whereas van den Berg et al. (2020) find positive employment effects of frequent meetings in the form of search clubs for youths in deprived neighborhoods in France.

year for immigrants than for other job seekers (the difference is insignificant if we apply the Bonferroni correction to adjust for multiple testing). Table A-4 reveals no significant differences in the effects of JSA across education groups. We also see that men benefit more than women in terms of direct effects, but also suffer more from displacement (not reported). These gender differences are in line with the results in Crépon et al. (2013).

6 How do displacement effects arise?

We now examine how displacement arises. We conduct a structural analysis of a full-scale roll-out of the program, analyze displacement and vacancy adjustment in the long run, and examine displacement in different types of labor markets. This may reveal if there are circumstances for which displacement is less important, which may help to target JSA to more favorable settings. A theoretical starting point is that conventional models of equilibrium unemployment with frictions do not generate displacement (see, e.g., Pissarides, 2000). In such models, if JSA increases search effort among treated job seekers, the number of jobs increase enough to absorb this increased search (labor supply), precluding displacement among the non-treated. To understand the displacement we therefore build on the theoretical models in Gautier et al. (2018) and Crépon et al. (2013), which incorporates job displacement in different ways. We use these two frameworks in different parts of this section without developing a unified theoretical framework.

6.1 Displacement of resources

Before analyzing the above-mentioned aspects of displacement we examine resource allocation at the active offices. If resources are spent on JSA for the treated, displacement could arise simply because resources are allocated away from the non-treated. This may be a larger concern when JSA programs and the regular services are financed by the same budget, as in our case and in Gautier et al. (2018), than when JSA is operated by private providers, as in Crépon et al. (2013). Although the JSA program we study was supposed to be fully funded by external funds, we cannot rule out displacement of resources. From a policy perspective, it is crucial to distinguish between displacement in the labor market and displacement through resource constraints. While expanding the JSA budget solves the problem with constrained PES resources, it is more challenging to come up with policies addressing displacement of jobs.

The results in Table 7 show no evidence of displacement of resources, as the estimates for “In a program area” are all close to zero and insignificant. The non-treated did not receive fewer meetings (Panel A columns 1–3) or vacancy referrals (Panel B columns 1–3), nor did they apply to fewer vacancy referrals (Panel B columns 4–6). They also met similar caseworkers (Panel A columns 4–7) in terms of caseload (number of meetings and clients per month) and tenure (tenure in days at the PES and the current office). We have also examined the other caseworker actions and search behavior measures used in Table 4 and found no evidence of any displacement effects. This suggests that the non-treated job seekers received the same level of assistance as they would have had in absence of the program. We conclude that the displacement in our experiment has little to do with reallocation of resources at the office level and that the displacement effects are due to displacement of jobs.

These results also provide additional insights on the vacancy referral mechanism. We have seen

that the referrals point the treated job seekers to the most relevant jobs earlier without changing the direction of search. Here, we see that the non-treated do not receive fewer referrals, which means that, on average, each vacancy gets more referrals. Since the program had no explicit aim to create new vacancies, this increased competition for the available jobs explain the sizeable displacement effects that we observe. But as examined below, if firms react to the improved search this may create new vacancies in the long run.²⁰

Table 7: Displacement of meetings and services

Panel A:	Displacement of meetings			Displacement of caseworkers			
	Meetings quarter 1	Physical meetings Q1	Distance meetings Q1	Meetings per month	Clients per month	Tenure at PES in days	Tenure at office in days
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assigned to program	0.498*** (0.052)	0.318*** (0.056)	0.180*** (0.047)	1.51 (1.59)	1.03 (1.26)	2.19 (7.10)	−6.94 (10.9)
In a program area	−0.019 (0.049) [3.38]	−0.053 (0.038) [2.51]	0.034 (0.023) [0.870]	−0.316 (4.94) [154.0]	1.03 (3.13) [104.0]	−16.6 (18.4) [1,186]	−7.58 (25.1) [946.5]
Clusters	72	72	72	72	72	72	72
Observations	552,816	552,816	552,816	552,732	552,732	552,732	552,732

Panel B:	Caseworker action: Vacancy referrals			Search behavior: Applications to vacancy referrals		
	Total	Mandatory	Suggested	Total	Mandatory	Suggested
	(1)	(2)	(3)	(4)	(5)	(6)
Assigned to program	0.111* (0.066)	0.031 (0.040)	0.080 (0.050)	0.097*** (0.035)	0.028 (0.018)	0.069** (0.027)
In a program area	−0.014 (0.081) [1.95]	0.012 (0.025) [0.263]	−0.026 (0.072) [1.68]	−0.020 (0.023) [0.397]	−0.004 (0.006) [0.045]	−0.016 (0.022) [0.351]
Clusters	72	72	72	72	72	72
Observations	552,816	552,816	552,816	135,961	135,961	135,961

Notes: Regression of each outcome variable on an indicator for active PES office (“In a program area”) and an indicator for active PES office $\times$ intention to treat status is treated (“Assigned to program”), weighted by the treatment share within each office (see footnote 6). Controls include the variables in Table 1, indicator variables for year, month and PES office, and randomization strata fixed effects. The sample includes active and non-active offices during 2012–2015. Standard errors clustered at the PES office level in parentheses, control means in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2 Structural evaluation of the equilibrium effects

We next examine a full-scale roll-out of the JSA program using an adapted version of the Diamond-Mortensen-Pissarides (DMP) model in Gautier et al. (2018). The model, described in detail in Appendix C, defines an equilibrium in which search intensity, wages, unemployment and labor market

²⁰We can also use the vacancy estimates and the displacement estimates to further study displacement of jobs. Table 4 shows that each non-treated job seeker gets on average 2.05 vacancy referrals per month and they apply to on average 7.81 jobs per month, so that the referrals constitute 26.2% of the job applications. The table also shows that the program increases the number of vacancy referrals by 5%. It implies that the job finding rate for the treated would increase by 1.3 percentage points (0.262×0.05) if the success rate is the same for referred and non-referred job applications, and by slightly more if the success rate is higher for the referred applications. If the number of vacancies is fixed and since the treated and non-treated groups are of roughly the same size, any higher job finding among the treated will be reflected in an equally large reduction of the job finding among the non-treated. This is consistent with the 1.5 percentage points reduction in job access for the non-treated we see in Table 3. The fact that the numbers add up provides some additional evidence in favor of the vacancy referral mechanism and its implications for displacement of jobs.

tightness are determined. Identical firms post vacancies, and the ex-ante identical job seekers choose the number of job applications they file, trading off job prospects and search costs. Job seekers only differ in whether they participate in the JSA program, and participation is allowed to affect, e.g., search costs and non-market time. To allow for job displacement, the model has an endogenous matching function, where the success of an application also depends on the number of applications sent by other job seekers. Compared to conventional search models, this is a direct way to create job displacement as any improved search among the treated will decrease the acceptance rate per job application for the non-treated.²¹

The full model estimates of the DMP model are reported in Table C-2 in Appendix C, and this table also indicates that the model fits the data well. One result is that the treated have a higher success rate per job application, which is consistent with the vacancy referrals directing the job seekers to the most relevant jobs. This creates search congestion, which decreases the job application acceptance rate for the non-treated, offering one explanation to the sizeable displacement that we document. When we simulate the impact of a full-scale roll-out, we see that the matching rates for both treated and non-treated are decreasing in the share of treated, since a larger share of treated creates more search congestion. Overall, however, the aggregate matching rate increases with the share of treated as the matching rate is higher for the treated than for the non-treated. We also see that wages are decreasing in the share of treated, because a higher share of treated creates more search congestion, which lowers the outside option for the job seekers.

The higher matching rate and the lower wages induce firms to create more vacancies, so that market tightness is increasing in the share of treated. Therefore, a full-scale roll-out, increasing the share of treated from 0 to 100 percent, would lower the unemployment rate by around 0.2 percentage points (see Table C-3 in Appendix C). It would also lead to a small reduction of government spending, as the decrease in benefit payments as a result of the lower unemployment rate is of roughly the same magnitude as the program costs (Section 2.1 describes program costs).²² However, a full-scale roll-out of the program would reduce welfare, for instance, because job seekers lose non-market time when participating in the program. All these simulation results are discussed in more detail in Appendix C.

6.3 Displacement in the long run

The structural analyses above assume that the market reached the new equilibrium during the experiment and that there will be no further adjustments of vacancies in the long run if the experiment was made permanent. However, a limitation with most JSA experiments is that they capture the short-run effects during the experiments, and Cahuc and Le Barbanchon (2010) show that transitory and permanent policies may have very different effects. Our JSA program targeted newly unemployed job seekers during six months, Crépon et al. (2013) sampled participants during a 13-month period, and Gautier et al. (2018) enrolled newly unemployed job seekers for four months. These short-run experiments may not account for the full effects if it takes time for firms to observe and react to the new economic environment. In particular, any improved search (and lower wages) should induce firms

²¹To capture the vacancy referral mechanism we make a minor extension of the model in Gautier et al. (2018), using a matching function with differential job application success rates for the treated and the non-treated.

²²Since taxes are paid on both labor earnings and the benefit payments, we abstract from any changes to tax receipts in the main analyses. Also, since wages are higher than benefits, taking tax receipt into account would make the positive impact on government spending slightly more positive (increase by around 9%) but leave the welfare calculations unchanged.

to open more vacancies, leading to an increased job finding rate for both the treated and the non-treated and, thus, more favorable employment effects of JSA. If these vacancy responses are delayed, the long-run impact of JSA may be different from the short-run impact during the experiment.

It is of course challenging to study long-run impacts of a short-run experiment. We examine whether there are indications of an increasing vacancy response as the experiment proceeds, which would provide some suggestive evidence of further adjustments of vacancies in the long run in case the program was made permanent. To study vacancy responses during the program period, we analyze the impact on the log flow and log stock of vacancies in the spring and fall of 2015, when the treated individuals were active job seekers.²³ The estimates in Table 8 indicate an interesting pattern of increasing vacancies at the active compared to the non-active offices during the second half of the JSA program. The flow (stock) of vacancies increases by around 9.0% (5.6%) during fall, but only the effect on the flow is significant. Since we use pre-experiment data to adjust for office heterogeneity, we rely on a parallel trends assumption in the vacancy rates for the active and non-active offices before the start of the experiment. Assuringly, Table A-5 in Appendix A reveals no significant pre-trends in the 10 months before the start of the experiment. However, we note that the size of these pre-trends are sizeable, suggesting that our main estimates should be interpreted with some caution. A delayed vacancy response would also imply that the displacement effects are smaller towards the end of the experiment. We find some suggestive evidence in this direction. Columns 3–5 of Table 8 show that displacement effects (“In a program area”) are smaller during the fall of 2015 than during the spring, but the differences are insignificant.

Table 8: Effects on vacancies and unemployment in the program areas, over time

	Vacancies		Unemployment		
	Log flow vacancies (1)	Log stock vacancies (2)	Exit unemp. 1st quarter (3)	Unemp. days 1st quarter (4)	Unemp. days 1st year (5)
Assigned to treatment, spring 2015			0.026** (0.012)	−1.027 (0.683)	−6.423** (2.569)
Assigned to treatment, fall 2015			0.038*** (0.005)	−1.718*** (0.388)	−5.729*** (1.799)
In a program area, spring 2015	0.022 (0.036)	0.005 (0.041)	−0.027** (0.010)	1.100** (0.527)	7.844*** (2.514)
In a program area, fall 2015	0.090*** (0.032)	0.056 (0.036)	−0.008 (0.007)	0.451 (0.370)	1.932 (2.083)
Clusters	72	72	72	72	72
Observations	4,320	4,320	552,816	552,816	552,816

Notes: Regression of the log flow and log stock of vacancies in each office and month on an indicator for active PES office (“In a program area”). Controls include indicator variables for year, month and PES office, and randomization strata fixed effects. Controls in columns 3–5 also include the variables in Table 1. The sample includes all active and non-active offices during 2012–2015. Standard errors clustered at the PES office level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The results provide some suggestive evidence of a delayed vacancy response, with further demand side responses and adjustment of vacancies if the program was made permanent. To explore the

²³We use a similar model as above, comparing vacancy rates across active and non-active offices during the experiment and adjusting for office heterogeneity using pre-experiment data back until 2012. The main difference is that we have one outcome measure per office and month and focus on the program area dummy, which takes the value one for the active offices during the experiment period.

implications of this we incorporate a delayed vacancy response into the structural model above. We estimate the model assuming that job seekers adjust their search but firms do not adjust vacancies during the experiment, and use the model to simulate the new equilibrium when vacancies have adjusted, aiming to capture the long-run impacts of the JSA program. The simulation results in Table C-3 in Appendix C predict larger reductions of unemployment and government expenditures than the baseline analyses in Section 6.2. Importantly, the negative welfare effects we see in the baseline analysis are now reversed to positive. It suggests that the overall assessment of JSA programs based on short-run experiments hinges on how broadly the displacement effects hit the labor market and to what extent firms react to the new conditions.²⁴ Therefore, exploring delayed vacancy responses is an interesting topic for future research.

6.4 Displacement in different types of labor markets

We next build on the theoretical framework in Crépon et al. (2013), which is a simplified version of Michaillat (2012) and Landais et al. (2018a,b), and examine displacement in different types of labor markets. If the size of displacement varies, targeting JSA to certain labor markets is one way to design more efficient JSA policies. Compared to conventional search models, this job-rationing framework assumes decreasing returns to scale in the production function. This implies that job creation does not fully adjust in response to increased search among the treated (increased labor supply), and the model therefore predicts that the non-treated are partly displaced by the treated.²⁵ For our purposes, a key prediction of the model is that displacement is larger and the overall effect of JSA more negative if labor demand is low than if demand is high. It follows the reasoning in Landais et al. (2018a,b), and relates to their argument that unemployment insurance often should be more generous in recessions. Crépon et al. (2013) also provide empirical evidence that JSA appears to be more effective when labor demand is low.

Columns 1–2 in Table 9 re-examines the effects of JSA when labor demand is high and low. In separate analyses we define high (low) demand as the local unemployment rate being below (above) the median rate and the number of vacancies per job seeker being above (below) the median rate. Consistent with Crépon et al. (2013), the results indicate that displacement is larger in slack labor markets (low demand) than in tight markets (high demand).²⁶ Our interpretation is that the vacancy information channel behind the direct effect works for both tight and slack labor markets, but the harm for the non-treated job seekers is limited under good labor market conditions, when there is no job rationing and the access to alternative jobs is good. In contrast, when competition for jobs is higher, the non-treated are affected more by the jobs obtained by the treated.

We are also interested in whether the impact of JSA depends on the search behavior in the market

²⁴In a related study, Cottier et al. (2018) compare treated and non-treated to provide valuable insights on the direct effects of JSA in the long run. We offer complementary evidence on the importance of vacancy responses and job displacement in the long run.

²⁵Formally, the job rationing model outlined in Crépon et al. (2013) is based on the idea that JSA increases search effort. Our setting is a bit different since our JSA program leads to more vacancy referrals and more focused search rather than increased search effort. However, note that the vacancy information means that the treated end up first in line for the available jobs, and using a job rationing argument this should lead to larger displacement effects (more negative effects on job finding rates for the non-treated) if labor demand is low and the available jobs are few.

²⁶One concern with the analysis in Table 9 is that the compliance rate (actual number of meetings) may correlate with local labor market conditions, but we see no clear correlation with the compliance rate nor with other labor market characteristics. Moreover, if the compliance rate differs, we expect to see systematic differences in the direct ITT effect for the treated across high and low demand markets, which we do not see.

before the JSA program. For instance, if search is low and there are large search frictions, employment may be constrained by job seekers' search behavior rather than by low labor demand. If JSA leads to a larger improvement in the matching of firms and job seekers for low search markets, it should induce firms to open more vacancies. JSA may then have more favorable overall employment effects for low search markets than for high search markets. We do not theoretically formalize these arguments, and therefore view our analyses as explorative, hopefully providing input to future research.

Table 9: Displacement effects of the JSA program in strong and weak labor markets on unemployment days during the 1st quarter

	Labor demand		Search
	Unemployment rate (1)	Labor market tightness (2)	Search intensity (3)
Low demand×Assigned	−1.588*** (0.592)	−2.674*** (0.840)	
High demand×Assigned	−1.416** (0.560)	−1.041** (0.408)	
Low demand×Program Area	1.413*** (0.521)	1.749** (0.724)	
High demand×Program Area	0.028 (0.468)	0.345 (0.366)	
Low search ×Assigned			−1.184** (0.468)
High search ×Assigned			−1.745*** (0.616)
Low search ×Program Area			0.367 (0.555)
High search ×Program Area			0.993** (0.476)
Clusters	72	72	72
Observations	552,816	552,816	552,816

Notes: Regression of unemployment days during the first quarter on an indicator for treatment PES office (“In a program area”) and an indicator for treatment PES office × intention to treat status is treated (“Assigned to program”) and both interacted with whether the municipality of the local PES office had low or high demand/search, weighted by the treatment share within each office (see footnote 6). The model also includes year dummies, month dummies and local PES dummies. Columns 1 and 2 in Panel A control for unemployment rate and labor market tightness, respectively. Low (high) demand is defined as the monthly unemployment rate being above (below) or labor market tightness being below (above) the median across Sweden. Low (high) search is defined as the average pre-experiment search intensity below (above) the median across Sweden. The sample includes all active and non-active offices during 2012–2015. Standard errors clustered at the PES office level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In column 3 in Table 9 we classify all local labor markets (offices) based on the initial search in the market before the experiment. Since we only observe the equilibrium outcome, it is difficult to separate between demand and search factors, but as an attempt we exploit individual job search information from the PES and define high search markets as having above median search intensity before the JSA program, according to job seekers' monthly activity reports.²⁷ The results point in the direction of smaller displacement effects for low search markets than for high search markets, but we note that the difference is insignificant. It does not rule out that the vacancy information improves the search in all types of markets, but this channel and its implications for job seekers and firms matching may

²⁷In a standard search model, an increase in the number of vacancies should result in increased search effort, so that our search and demand measures should be correlated. This is also what we see in the data, but the correlation is rather low, and using search intensity leads to a different split of the local markets than splitting on local demand.

be more important for markets constrained by low search.

We have also examined whether the impact on the search behavior that was induced by the JSA program—the number of vacancy referrals applied to—differ by local labor market conditions, but Table A-6 in Appendix A reveals no systematic patterns. Note that this does not contradict our previous results, but it has some implications for the interpretation of our results. One is that the more positive overall effects of JSA for high demand markets is explained by less search congestion per referral rather than by differences in induced search by the program.

7 Conclusions

This paper has evaluated a large-scale two-level randomized experiment on JSA that added extra meetings with a caseworker during the first quarter of unemployment. In line with the previous literature on job search assistance, we find both positive direct effects for the treated and substantial negative displacement effects for the non-treated. By exploring unique administrative data on caseworkers' actions and job seekers' search behavior we have been able to shed further light on the mechanisms behind our results. We show that vacancy referrals from caseworkers to job seekers are a key component behind the positive effects for the treated. Caseworkers pinpoint job seekers to relevant job openings and job seekers apply to these positions. The vacancy information does not lead to broader search, but rather helps job seekers to apply to the most relevant jobs earlier. Given the low cost of our JSA program, our conclusion is that JSA with a strong focus on vacancy referrals can improve the job search process. Due to the randomization design, we also compared different meeting formats. We find equally large positive effects for face-to-face and distance meetings, strengthening the case for communication technologies when providing JSA.

We have also documented how these direct effects created by the vacancy referrals translate to displacement effects. Using detailed data we can rule out that the displacement is an artifact of the experiment due to crowding out of resources at the local PES offices. Instead, increased competition for the available jobs created by the vacancy referrals given to the treated job seekers explains the sizeable displacement effects that we observe. The displacement of jobs also has implications for JSA under different labor market conditions. Treated job seekers benefit from the JSA program in both tight and slack labor markets, but the harm for non-treated job seekers is limited under good labor market conditions when there are many available jobs. These labor market responses also have implications for vacancy creation and the long-run impacts of JSA programs. We find some evidence of larger changes in the vacancy rate towards the end of the experiment, which is consistent with a delayed vacancy response. Using a structural model we show that this implies more positive long-run effects when all demand side responses have occurred than what is captured during a temporary program.

These new results on vacancy information, meeting formats, displacement in different markets and vacancy adjustments in the long run can help policymakers to design more efficient JSA programs. Providing personalized vacancy information can be a key component of successful JSA—in particular if displacement can be limited for non-participants, for instance by targeting JSA to strong labor markets.

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Appendix A: Additional Tables and Figures (for online publication only)

Table A-1: Direct and displacement effects (ITT) of the JSA program on unemployment

	Experiment period in 2015			2012–2015	
	(1)	(2)	(3)	(4)	(5)
Panel A: Unemployment days 1st quarter					
Assigned to program	−1.574*** (0.343)	−1.509*** (0.331)	−1.518*** (0.330)	−1.505*** (0.399)	−1.456*** (0.353)
In a program area			0.849 (0.546)	0.725** (0.344)	0.695** (0.318)
Net effect treated			−0.669 (0.543)	−0.780** (0.334)	−0.761** (0.326)
Control mean	74.37	74.37	73.51	73.78	73.78
Panel B: Unemployment days 1st year					
Assigned to program	−6.692*** (1.628)	−5.929*** (1.503)	−5.975*** (1.303)	−6.602*** (1.778)	−5.991*** (1.359)
In a program area			5.687** (2.660)	4.318** (1.912)	4.160** (1.727)
Net effect treated			−0.288 (2.578)	−2.284 (1.612)	−1.831 (1.484)
Control mean	196.1	196.1	190.7	187.0	187.0
Year dummies	No	No	No	Yes	Yes
Month dummies	No	No	No	Yes	Yes
PES office dummies	No	No	No	Yes	Yes
Controls	No	Yes	Yes	No	Yes
Clusters	No	No	72	72	72
Observations	26,538	26,538	57,778	552,816	552,816

Notes: Regression of each outcome variable on an indicator for active PES office (“In a program area”) and an indicator for active PES office × intention to treat status is treated (“Assigned to program”), weighted by the treatment share within each office (see footnote 6). Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, clustered at the PES office level in columns 3–6. *** p<0.01, ** p<0.05, * p<0.1.

Table A-2: Impact outside target population

	Non-experiment periods 2015			Non-target population		
	Exit unemp. 1st quarter (1)	Unemp. days 1st quarter (2)	Unemp. days 1st year (3)	Exit unemp. 1st quarter (4)	Unemp. days 1st quarter (5)	Unemp. days 1st year (6)
In a program area	−0.009 (0.008)	0.699* (0.419)	1.585 (2.188)	0.011 (0.007)	−0.727 (0.581)	−3.049 (1.973)
Control mean	0.390	73.78	187.0	0.535	52.34	162.7
Clusters	72	72	72	72	72	72
Observations	552,816	552,816	552,816	367,778	367,778	367,778

Notes: Regression of each outcome variable on an indicator for active PES office (“In a program area”) during 2015. The regressions include year dummies, month dummies, PES office dummies, the control variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, clustered at the PES office level. *** p<0.01, ** p<0.05, * p<0.1.

Table A-3: Effects of being assigned to the JSA program on job quality

	Log wage (1)	Re-unemployed within 5 months (2)
Assigned to program	−0.001 (0.005)	−0.001 (0.007)
Control mean	10.13	0.144
Observations	8,577	12,227

Notes: Regression of each outcome variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes the active PES offices during the experiment period. *Log wage* for first job after unemployment, *Re-unemployed within 5 months* after first exit from unemployment. Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses.

Table A-4: Heterogenous effects of the being assigned to JSA program

	No college (1)	College (2)	Born in west EU (3)	Born outside west EU (4)
Exit unemployment 1st quarter	0.039*** (0.007) [0.362]	0.024** (0.011) [0.335]	0.030*** (0.007) [0.409]	0.050*** (0.010) [0.216]
Unemployment days 1st quarter	−1.730*** (0.390) [74.25]	−0.921 (0.626) [74.68]	−1.269*** (0.408) [71.84]	−2.228*** (0.540) [80.75]
Unemployment days 1st year	−6.083*** (1.763) [196.3]	−5.435* (2.854) [195.5]	−4.193** (1.748) [175.5]	−11.280*** (2.884) [248.1]
Observations	18,986	7,552	19,092	7,446

Notes: Regression of each outcome variable on a treatment indicator, weighted by the treatment share within each office (see footnote 6). The sample includes the active PES offices during the experiment period. Controls include the variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, control means in square brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table A-5: Effects on vacancies in the program areas, over time

	Log flow vacancies (1)	Log stock vacancies (2)
May–Sep 2014	0.038 (0.028)	0.041 (0.041)
Oct 2014–Feb 2015	−0.017 (0.033)	−0.069 (0.043)
Mar–July 2015	0.020 (0.039)	−0.011 (0.042)
Aug–Dec 2015	0.088** (0.036)	0.039 (0.041)
Jan–May 2016	−0.028 (0.034)	−0.070 (0.050)
Clusters	72	72
Observations	4,320	4,320

Notes: Regression of the log flow and log stock of vacancies in each office and month on an indicator for active PES office (“In a program area”). Controls include indicator variables for year, month and PES office, and randomization strata fixed effects. The sample includes all active and non-active offices during 2012–2015. Standard errors clustered at the PES office level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A-6: Displacement effects of the JSA program in strong and weak labor markets on vacancy referrals applied to during the 1st quarter

	Labor demand		Search
	Unemployment rate (1)	Labor market tightness (2)	Search intensity (3)
Low demand×Assigned	0.125** (0.057)	0.076 (0.051)	
High demand×Assigned	0.069** (0.034)	0.109** (0.045)	
Low demand×Program Area	−0.022 (0.035)	−0.070** (0.028)	
High demand×Program Area	−0.033 (0.030)	−0.009 (0.028)	
Low search ×Assigned			0.052 (0.041)
High search ×Assigned			1.131** (0.051)
Low search ×Program Area			−0.043 (0.029))
High search ×Program Area			−0.001 (0.034)
Clusters	72	72	72
Observations	135,961	135,961	135,961

Notes: Regression of vacancy referrals applied to in the first quarter on an indicator for treatment PES office (“In a program area”) and an indicator for treatment PES office × intention to treat status is treated (“Assigned to program”) and both interacted with whether the municipality of the local PES office had low or high demand/search productivity, weighted by the treatment share within each office (see footnote 6). The model also includes year dummies, month dummies and local PES dummies. Columns 1 and 2 in Panel A control for unemployment rate and labor market tightness, respectively. Low (high) demand is defined as the monthly unemployment rate being above (below) or labor market tightness being below (above) the median across Sweden. Low (high) search is defined as the average pre-experiment search intensity below (above) the median across Sweden. The sample includes all active and non-active offices during 2012–2015. Standard errors clustered at the PES office level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Appendix B: IV estimates of program participation (for online publication only)

The main analysis focuses on the ITT effect of the JSA program, but here we also report IV estimates of the LATE of program participation, defined as being informed about the program, i.e., by observing that the experiment-specific randomization tool had been used and the revealed treatment status was treated (see Table 2). The interpretation of the LATE depends on how representative program participants are compared to the target population as a whole, and Table B-1 shows that program participants are similar to the target population in terms of observable characteristics. This is expected since the dropout mainly occurred at the caseworker and the office level, and not because job seekers selected in or out of the program.

Table B-1: Sample statistics by treatment status

Variables	Assigned to program (1)	Program participant (2)
Age	33.33	34.34
Male	0.542	0.538
Unemployment benefits	0.642	0.692
Disabled	0.052	0.048
Matchable	0.868	0.884
Education level		
Less than high school	0.224	0.206
High school	0.491	0.494
College	0.285	0.300
Place of birth		
Sweden	0.678	0.700
Nordic countries	0.013	0.013
West Europe	0.036	0.036
Outside west Europe	0.273	0.251
Unemployment days		
Year t-1	30.66	31.30
Year t-2	67.42	69.16
Year t-3	69.57	73.08
Year t-4	63.82	67.95
Unemployment spells		
Year t-1	0.431	0.419
Year t-2	0.789	0.787
Year t-3	0.806	0.830
Year t-4	0.706	0.752
No. spells, last 4 yrs		
Labor market education	0.024	0.025
Preparatory education	0.048	0.047
Labor market training	0.027	0.027
Subsidized employment	0.106	0.122
Observations	14,075	8,358

Notes: Summary statistics, weighted by the treatment share within each office (see footnote 6). *Program participants* are treated job seekers for which the experiment-specific randomization tool was used. *Unemployment benefits* is an indicator variable for collecting unemployment benefits, *Health disability* is an indicator variable for having a functional disability and *Matchable* is an indicator variable from an initial assessment about the job seeker's potential to take a job on short notice.

To obtain the direct program effect, we instrument program participation with the program assignment indicator, $1(\text{Assigned to program}_{ij})$, and estimate:

$$Y_{ij} = \alpha_2 + \beta_3 1(\text{Program participation}_{ij}) + \beta_4 1(\text{In a program area}_j) + \varepsilon_{ij}, \quad (4)$$

where β_3 reflects the direct program effect at the active offices.

To estimate the net program effect for the participants, we follow the IV approach in Crépon et al. (2013). Then, the IV scaling takes into account that the compliers experience positive program effects, whereas the non-compliers experience negative displacement effects. The main identifying assumption is that the displacement effects are uncorrelated with treatment status. In practice, this means that compliers and non-compliers on average should have similar potential outcomes under no treatment. In our case, the compliers and the non-compliers have similar observed characteristics, lending support to this assumption. See Crépon et al. (2013) for details.

For comparison, columns 1, 3 and 5 of Table B-2 report the ITT effects from the main analyses for the three different outcome variables. Column 1 shows that the net effect on the exit from unemployment during the 1st quarter for treated job seekers is 1.8 percentage points, or 4.6%. By construction, the IV estimates in column 2 are larger than these ITT effects, with a net program effect corresponding to a 3.9 percentage point (10%) increase in the exit rate during the first quarter. The results for days in unemployment during the first quarter and the first year in columns 4 and 6 display similar patterns.

We can also compare our results to those in Schiprowski (2020) who finds that job seekers who lose a meeting with their caseworker stay unemployed about 5% longer. Our IV estimate for program participation with three extra meetings, not taking displacement into account, shows that the probability of leaving unemployment during the first quarter is 0.055 percentage points, or about 14.1% when compared with the control mean (column 2, Table B-2 in Appendix B).

Table B-2: ITT and IV effects of the JSA program on unemployment

	Exit unemployment 1st quarter		Unemployment days 1st quarter		Unemployment days 1st year	
	ITT (1)	IV (2)	ITT (3)	IV (4)	ITT (5)	IV (6)
Assigned to program	0.034*** (0.006)		-1.456*** (0.353)		-5.991*** (1.359)	
Program participant		0.055*** (0.010)		-2.352*** (0.570)		-9.679*** (2.195)
In a program area	-0.015** (0.006)	-0.015** (0.006)	0.695** (0.318)	0.695** (0.318)	4.160** (1.727)	4.160** (1.727)
Net effect treated ¹	0.018*** (0.005)	0.039*** (0.011)	-0.761** (0.326)	-1.657*** (0.552)	-1.831 (1.484)	-5.519 (2.936)
Control mean	0.390	0.390	73.78	73.78	187.0	187.0
Clusters	72	72	72	72	72	72
Observations	552,816	552,816	552,816	552,816	552,816	552,816

Notes: Regression of each outcome variable on an indicator for active PES office ("In a program area") and an indicator for active PES office $\times$ intention to treat status is treated ("Assigned to program"), weighted by the treatment share within each office (see footnote 6). All regressions include year dummies, month dummies, PES office dummies, the control variables in Table 1 and randomization strata fixed effects. Robust standard errors in parentheses, clustered at the PES office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

¹ Net effect in columns 2, 4 and 6 are for program participants.

Appendix C: Equilibrium model (for online publication only)

This appendix studies the implications of a full-scale roll-out of the program on labor market outcomes, government expenditures and welfare. To this end, we adapt the Diamond-Mortensen-Pissarides (DMP) model in Gautier et al. (2018) to our setting.

C.1 The model

The model defines an equilibrium in which search intensity, wages, unemployment and labor market tightness are determined. It is a discrete-time model with ex ante identical and risk neutral workers with the same productivity, who only differ in whether or not they participate in the JSA program (indexed by 0 or 1). The unemployed receive benefits b and the value of non-market time is h . In each period, the worker decides the number of job applications to file, a , trading off job prospects and search costs. For convenience, we assume that the search costs are quadratic in the number of applications (γa^2). If an application is successful the worker becomes employed, otherwise (s)he has to apply again in the next period.

A key part of the model is the endogenous matching function that captures that the success of an application depends on the number of applications sent by other workers. The matching rate for a worker who sends a applications, $m(a; a_0, a_1, \tau, \theta)$, is specified below. Since we will allow for differential success rates for participants and non-participants, this matching rate depends separately on the number of application sent by the program participants, a_1 , the number of application sent by non-participants, a_0 , and the share of treated, τ . It further depends on labor market tightness, $\theta \equiv v/u$, where u is the unemployment rate and v the vacancy rate.

Let r be the discount rate, and $E(w)$ the value of being employed at the wage rate w . Then, the value of unemployment for a non-participant is²⁸

$$rU_0 = \max_{a \geq 0} b + h - \gamma_0 a^2 + m_0(a; a_0, a_1, \tau, \theta) [E(w_0) - U_0]. \quad (5)$$

From the first order condition we have that the optimal number of applications is:

$$a_0^* = \frac{E(w_0) - U_0}{2\gamma_0} \frac{\partial m_0(a; a_0, a_1, \tau, \theta)}{\partial a} \Big|_{a=a_0^*}. \quad (6)$$

The participants receive more meetings with caseworkers through the JSA program. We allow this to have three separate effects. First, obtaining job search assistance may reduce the participants' search costs, γ_1 , in relation to the costs of non-participants, γ_0 . Second, participating in the program costs non-market time. We normalize its value to zero for participants and let it be h for the non-participants. In the estimations, we do not restrict the sign of h , even though our prior is that the value of non-market time is lower for the participants. Third, we specify separate matching functions for participants and non-participants under the assumption that the treatment may affect job-search efficiency and the success rate per job-application (see below). Thus, the value of unemployment for

²⁸Note that this expression implicitly means that benefits and search costs are realized at the end of the period. This simplifies the notation.

a program participant is

$$rU_1 = \max_{a \geq 0} b - \gamma_1 a^2 + m_1(a; a_0, a_1, \tau, \theta) [E(w_1) - U_1], \quad (7)$$

and the first order condition gives

$$a_1^* = \frac{E(w_1) - U_1}{2\gamma_1} \frac{\partial m_1(a; a_0, a_1, \tau, \theta)}{\partial a} \Big|_{a=a_1^*}. \quad (8)$$

The value of employment depends on the wage, w , the exogenous job destruction rate, δ , and the difference between the flow values of employment and unemployment:

$$rE(w_i) = w_i - \delta[E(w_i) - \bar{U}] = w_i - \delta[E(w_i) - U_0], \quad (9)$$

where $\bar{U} = \tau U_1 + (1 - \tau)U_0$ is the average utility.²⁹

Firms are also assumed to be identical. The value of a vacancy, V , for a firm is determined by the vacancy cost, c_v , the probability of filling the vacancy, $\frac{m(a_0^*, a_1^*, \tau, \theta)}{\theta}$, and the value difference between a filled, $J(w)$, and an unfilled vacancy:

$$rV = -c_v + \frac{m(a_0^*, a_1^*, \tau, \theta)}{\theta} (J(w) - V). \quad (10)$$

The value of a filled vacancy is given by

$$rJ(w) = y - w - \delta(J(w) - V), \quad (11)$$

where y is the value of output in each period. With free entry, the value of a vacancy is zero in equilibrium. From (10), (11) and this zero-profit condition we have

$$\frac{m(a_0^*, a_1^*; \tau, \theta)}{\theta^*} = \frac{(r + \delta)c_v}{y - \bar{w}^*}, \quad (12)$$

which can be solved for the equilibrium value of tightness, θ^* , since the left-hand side is decreasing in θ and the right-hand side is increasing in θ , so that there is a unique θ^* that satisfies equation (12).

We now specify the matching functions for participants and non-participants, allowing for different search intensities and different success rates per application for the two groups. We use an urn-ball model, implying that a firm receiving many applications randomly selects one and declines the others. This creates search congestion, since the probability that an application is accepted depends on the number of applications sent by other workers. Note that the empirical analyses in earlier sections show that the JSA program leads to more vacancy referrals and a higher job-finding rate, without changing the total number of job applications. This implies a higher probability that an application results in a job offer for the participants than for the non-participants. To incorporate this, we allow the job applications from the participants to have a higher probability of being drawn from the pool of candidates than the applications from non-participants.

²⁹Workers presumably realized that the JSA program was temporary. We therefore replace $\bar{U}$ in (9) by U_0 when estimating the model, but use $\bar{U} = \tau U_1 + (1 - \tau)U_0$ in the policy simulations, assuming that workers expect to be non-participants if they re-enter unemployment after losing their job.

Specifically, we introduce a parameter, ω , such that an application from a program participant results in a job offer with probability $\frac{\omega}{\omega + \omega j_1 + j_0}$, if the number of participant competitors equals j_1 and the number of non-participant competitors for the job is j_0 . For non-participants this probability is $\frac{1}{1 + \omega j_1 + j_0}$. By allowing both search costs and the success rate per application to differ, the model shows to what extent the higher job-finding rate for the program participants is due to changes in the search costs, or due to more suitable job applications (higher success rate per application).

If the number of workers and vacancies are sufficiently large, the number of applications from participants and non-participants are approximately Poisson distributed random variables with means $\tau a_1^*/\theta$ and $(1 - \tau)a_0^*/\theta$, respectively. Thus, for the non-participants the probability that an application results in a job offer is:

$$\psi_0 = \sum_{j_1=0}^{\infty} \sum_{j_0=0}^{\infty} \frac{1}{1 + \omega j_1 + j_0} f_0(j_0) f_1(j_1), \quad (13)$$

where $f_0(j_0) = \frac{\exp(-[1-\tau]a_0^*/\theta)([1-\tau]a_0^*/\theta)^{j_0}}{j_0!}$ and $f_1(j_1) = \frac{\exp(-\tau a_1^*/\theta)(\tau a_1^*/\theta)^{j_1}}{j_1!}$ are the probability of j_1 applications from participants and j_0 applications from other non-participants.

For participants, we have

$$\psi_1 = \kappa \sum_{j_1=0}^{\infty} \sum_{j_0=0}^{\infty} \frac{\omega}{\omega + \omega j_1 + j_0} f_0(j_0) f_1(j_1). \quad (14)$$

Here, the parameter κ captures other effects of the JSA program on job search efficiency. For example, the program may affect the type of jobs the participants apply to, thereby affecting the share of vacancies for which a participant has a positive productivity. The idea is that the firm first selects a worker for the job and then learns whether the worker is productive or not.³⁰

All this leads to the matching functions $m_0 = 1 - (1 - \psi_0)^{a_0^*}$ and $m_1 = 1 - (1 - \psi_1)^{a_1^*}$, and the aggregate matching function $m(a_0^*, a_1^*, \tau, \theta) = \tau m_1 + (1 - \tau)m_0$.

Wages are set in a Nash bargaining when workers and firms have met, with worker bargaining power equal to β . Since participants and non-participants have different outside options, U_1 and U_0 , we allow their equilibrium wages to differ. The bargaining outcome is given by

$$w_i^* = \arg \max_{w_i} [E(w_i) - U_i]^\beta [J(w_i) - V]^{1-\beta}, \quad (15)$$

and the first order condition gives

$$(1 - \beta) [w_i^* + \delta \bar{U} - (r + \delta)U_i] = \beta [y - w_i^*]. \quad (16)$$

In equilibrium, inflow into and outflow from unemployment are equal, and, hence, equilibrium unemployment is

$$u^* = \frac{\delta}{\delta + m(a_0^*, a_1^*, \tau, \theta)}. \quad (17)$$

The equilibrium can now be defined as $\{a_0^*, a_1^*, \theta^*, w_0^*, w_1^*, u^*\}$ satisfying equations (6), (8), (12), (16), and (17).

³⁰With different success rates per application for the two groups via ω one may argue that κ is redundant. Since κ is included in the Gautier et al. model we include it for completeness, but note that this has little impact on our estimates.

C.2 Estimation

We estimate the model using the actual share of program participants in the eligible population. The program participants corresponds to 62% of those assigned to the program. With a fraction assigned to the program of about 0.5, the share of program participants during the experiment, τ^e , is 0.31. Note that this share is based on the participants and non-participants included in the experiment. The non-participants also include workers in the stock of unemployed before the experiment, and workers not in the target population (immigrants and workers with repeated spells of unemployment). These groups were not included in the experiment, and our analyses revealed no evidence of any displacement effects for these groups, indicating that the displacement mainly occurs within the target population. The absence of displacement outside the experiment explains why we use the share of participants in the target population. But, below we also report results from sensitivity analyses using lower treatment shares, implicitly allowing for displacement effects for the non-target population.

Estimation is based on indirect inference using the reduced form estimates from the experiment. The model estimates are then used to simulate a full-scale roll-out of the program. The data moments used in the estimations, adapted to the monthly intervals in the discrete-time model, are displayed in Table C-1 in the appendix. At the individual level, we use estimates for the effect of program participation and estimates of the displacement effect on the exit rate from unemployment from Table B-2. As already mentioned, IV estimates are used since they reflect the program effects for the participants. We have also estimated similar program participation and displacement effects for wages and the number of program applications, corresponding to the IV and displacement version of the wage estimates in Table A-3 and the job application estimates in Panel B of Table 4. Using information on job applications helps to capture if the treatment is due to changed search costs or changed success rate per application. We also use estimates for the average effect on the vacancy rate for the full experimental period.

The data moments also include the average re-employment rate from the PES data, the average vacancy rate, the average number of job applications, the unemployment rate facing the target population in each office based on their municipality of residence, and the average replacement rate (fraction of previous income replaced by the unemployment insurance) from calculations by the National Institute for Economic Research (2016). More details are presented in Table C-1. Finally, the interest rate is set to $r = 0.008$, which is the monthly interest rate implied by an annual rate of ten percent, and we normalize productivity to $y = 1$.

C.3 Government expenditure and welfare

Besides effects on unemployment, wages and labor market tightness, we study government expenditure and welfare. The expenditures include unemployment benefits, b , and program costs, c_p , per worker. Then, government spending as a function of the share of participants is

$$C_B(\tau) = ub + \delta(1 - u)\tau c_p, \quad (18)$$

where ub is the fraction of unemployed times the unemployment benefits, and $\delta(1-u)\tau$ is the fraction of new entrants into unemployment (job destruction rate, δ , times the employment rate, $(1-u)$).³¹

To estimate the costs, we use the time-use survey described in Section 2.1. It shows that caseworkers spent on average 75 minutes per meeting and 225 minutes in total for the three extra meetings in the program, which corresponds to 2.5% of the monthly working hours. Next, based on calculations made by the PES, we assume that the average caseworker wage equals the average wage obtained by the workers in our target population. With the monthly production, y , normalized to one, the average wage is 0.96. We also add overhead costs for premises, administration and managers, estimated to roughly 30%. In total, the estimated program cost is 0.026 of monthly production.

The welfare effects of the program summarize a number of distinct elements. First, we have the resources spent on the program, c_p , assumed to be funded by a non-distortionary tax to avoid complications introduced by tax incentives and their effects on job search behavior. Second, we have an effect on output, $(1-u)y$, of employment changes induced by the program. Third, we have vacancy costs, vc_v , which vary with the number of open vacancies, v , and the cost per vacancy, c_v . Fourth, both participants and non-participants experience search costs as a function of γ_1, γ_0 and the number of applications. Finally, program participation implies a loss of non-market time, h . In total, welfare, $W(\tau)$, is given by

$$W(\tau) = (1-u)y + u \left((1-\tau) \frac{h - \gamma_0 a_0^{*2}}{1+r} + \tau \frac{-\gamma_1 a_1^{*2}}{1+r} \right) - \delta(1-u)\tau c_p - vc_v. \quad (19)$$

³¹This incorporates the costs of the program, but if the program leads to lower unemployment it will also reduce the costs for the baseline services given to both participants and non-participants. Here, we abstract from this channel. One reason is that it is difficult to obtain reliable costs measures for the baseline services.

Table C-1: Reduced form estimates used in the equilibrium search model (moment conditions)

	Data moment and inverse weight	Description	Value in the model
Panel A: Estimated program effects			
Exit rate, treated	0.039 (0.011 ²)	IV estimates for log wage (see Table B-2)	$[1 - (1 - (m_1 \tau = \tau^e))^3] - [1 - (1 - (m_0 \tau = 0))^3]$
Exit rate, non-treated	-0.015 (0.006 ²)	Displacement effect (see Table B-2)	$[1 - (1 - (m_0 \tau = \tau^e))^3] - [1 - (1 - (m_0 \tau = 0))^3]$
Log wage, treated	-0.002 (0.008 ²)	IV estimates for log wage	$\frac{w_1 \tau = \tau^e - w_0 \tau = 0}{w_0 \tau = 0}$
Log wage, non-treated	-0.0004 (0.005 ²)	Displacement effect for log wage	$\frac{w_0 \tau = \tau^e - w_0 \tau = 0}{w_0 \tau = 0}$
Log vacancy rate	0.031 (0.018 ²)	Average vacancy rate as in Table 8	$\frac{v^* \tau = \tau^e - v^* \tau = 0}{v^* \tau = 0}$
Job applications, treated	-0.181 (0.17 ²)	IV estimate for job applications	$a_1 \tau = \tau^e - a_0 \tau = 0$
Job applications, non-treated	-0.107 (0.10 ²)	Displacement effect for job applications	$a_0 \tau = \tau^e - a_0 \tau = 0$
Panel B: Aggregate statistics			
Exit rate (three months)	0.37 (0.003 ²)	Fraction of job seekers in treated offices who exit for employment within three months.	$1 - \tau(1 - (m_1 \tau = \tau^e))^3 - (1 - \tau)(1 - (m_0 \tau = \tau^e))^3$
Unemployment rate	0.077 (0.0001)	Unemployment rate treated offices.	$u^* \tau = \tau^e$
Vacancy rate	0.011 (0.001)	Vacancy rate treated offices. Calculated using information on vacancies at the municipal level.	$v^* \tau = \tau^e$
Replacement rate	0.60 (0.001)	Average replacement rate in Sweden.	$b \frac{1}{w^* \tau = 0}$
Job applications	3.46 (0.052 ²)	Average number of job applications in treated offices.	$\bar{a} \tau = \tau^e$

Notes: Empirical moments used in the estimation of the equilibrium search model. The interest rate is set to 0.008, which is the monthly interest rate implied by an annual rate of ten percent, and we normalize productivity to 1. Panel A reports moments from the estimated program effects with variances estimated in the empirical models. The estimates for log wages and job applications are IV effects for the effect of program participation and displacement effects using the estimation approach in Appendix B. Panel B reports moments in the form of aggregate statistics for the treated areas. For exit rate and job applications variances across unit are used. Unemployment rate and replacement rate do not have estimated variances. Here, the variances are set to capture that the unemployment rate is measured precisely, whereas the vacancy rate and replacement rate are less precisely estimated.

C.4. Results

Table C-2: Model fit and model estimates for the equilibrium search models

Panel A: Model fit		
	Moment	Deviation from the moments
<i>Estimated program effects</i>		
Exit rate, treated	0.039	-0.001
Exit rate, non-treated	-0.015	-0.001
Log wage, treated	-0.002	-0.001
Log wage, non-treated	-0.0004	-0.001
Log vacancy rate	0.031	0.020
Job applications, treated	-0.181	0.007
Job applications, non-treated	-0.107	0.111
<i>Aggregate statistics</i>		
Exit rate (three months)	0.37	0.007
Unemployment rate	0.077	0.000
Vacancy rate	0.011	0.014
Replacement rate	0.60	0.002
Job applications	3.46	-0.001
Panel B: Model estimates		
	Parameter	Estimates
<i>Fixed parameters</i>		
Treatment share	τ^e	0.31
Discount rate	r	0.008
Productivity	y	1
<i>Estimated parameters</i>		
Application cost, non-treated	γ_0	0.017 (0.006)
Application cost, treated	γ_1	0.020 (0.011)
Job destruction rate	δ	0.024 (0.004)
Share of vacancies w. pos. surplus	κ	0.920 (0.164)
UI benefits	b	0.577 (0.095)
Bargaining power	β	0.586 (0.199)
Vacancy costs	c_v	1.107 (1.168)
Value of non-market time	h	0.044 (0.0474)
Return to application, treated	ω	1.293 (0.861)

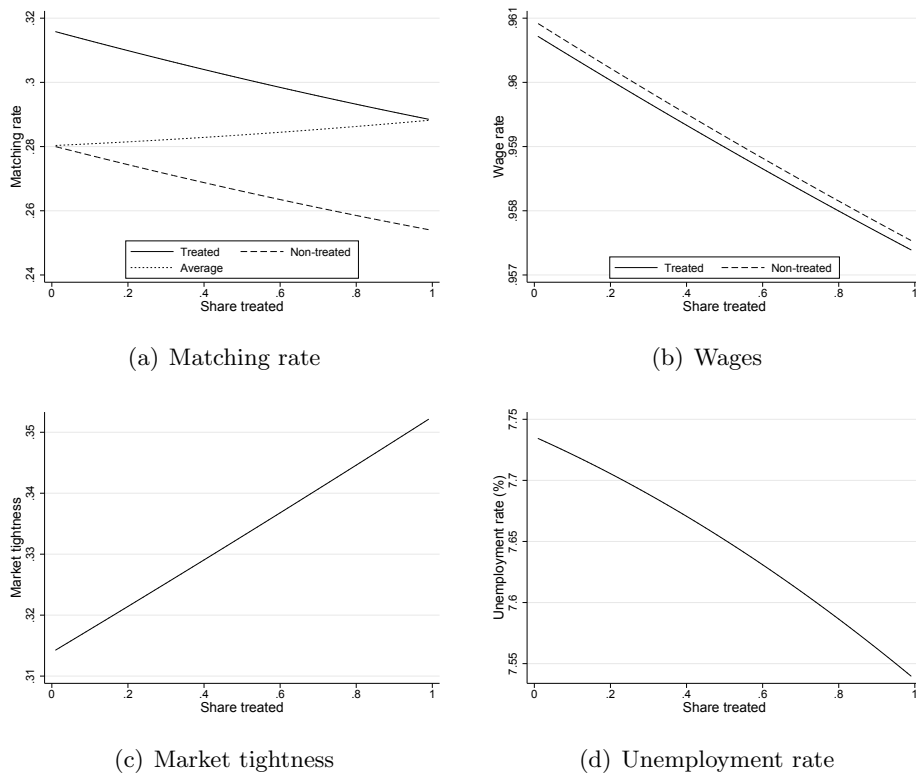
Notes: The table summarizes the model fit and presents the model estimates.

We now turn to the simulation results. Initially, Panel A of Table C-2 in the appendix assesses the model fit, displaying the difference between the moments implied by the model and the data moments. Generally, the fit is good: the program effects on the job-finding rate and the wage rate all match perfectly, as do the aggregate statistics. Panel B of Table C-2 presents the model estimates. Let us comment on some of the estimates. The estimate of ω implies that the participants have a higher probability to be drawn from the pool of candidates, leading to a higher success rate per application. Participants also have higher search costs than non-participants ($\gamma_1 > \gamma_0$). The higher success rate per application combined with the increased search costs imply that participants apply to roughly the same number of jobs as non-participants. This is consistent with the analyses of the mechanisms, which suggest that program participants receive more vacancy referrals, leading to a higher job-finding rate even though they do not apply for more jobs in total.

Figure C-1 presents simulation results for different shares of participants, including a full-scale roll-out. Recall that the reduced form estimates show that the participants find jobs faster, and that the non-participants find jobs at a lower rate due to displacement of jobs. The simulation results are

qualitatively similar, with a matching rate almost four percentage points, or around 15%, higher for participants compared to non-participants. We also see that the matching rate for both participants and non-participants is decreasing in the treatment share, since a larger share of participants creates more search congestion. Overall, however, the aggregate matching rate increases with the share of program participants. Figure C-1 also shows that participants receive somewhat lower wages than non-participants, and that wages for both groups are decreasing in the treatment share. The latter is because more participants create more search congestion, which lowers the outside options of all workers. The higher matching rate and the lower wages induce firms to create more vacancies, so that market tightness is increasing in the share of participants.

Figure C-1: Simulation results for different shares of participants



Notes: Outcomes by share of treated. Monthly matching rate with average weighted by the share of treated. Wages normalized with the monthly output per worker set to 1.

Overall, the results in Figure C-1 imply that unemployment decreases with the share of program participants. Increasing the share of participants from 0 to 100 percent lowers the unemployment rate by around 0.2 percentage points, implying that the net effect of the JSA program on unemployment is positive. Next, column 1 in Table C-3 summarizes the impact on unemployment, and evaluates the impact on government expenditure and welfare. We see that the JSA program reduces government spending, and that a full-scale roll-out of the program gives the lowest costs. Besides the case with monthly productivity normalized to one, the change in government expenditures is also calculated under the assumption that the monthly productivity is SEK 25,000 (average wage rate for the target population). These calculations show that a full-scale roll-out would decrease government expenditures by SEK 14 ($\approx$ 1.4 euros) per worker, i.e., a small effect. This means that the decrease

in benefit payments as a result of the lower unemployment rate is of roughly the same magnitude as the direct program costs.

We next study the welfare implications, which also take changes to search costs, non-market time, and vacancy costs into account. The simulation results in column 1 in Table C-3 reveal that welfare is decreasing in the share of participants. Thus, the positive effect on production from the decreased unemployment rate cannot compensate for the decreased value of non-market time, the direct program costs, and the increased vacancy costs induced by the increased vacancy rate.

Table C-3: Model predictions for the equilibrium search model

	Main model	Delayed vacancy model	Sensitivity analyses Treatment share 20%
	(1)	(2)	(3)
<i>Experiment ($\tau = 0.31$)</i>			
Unemployment (measured in %)	−0.047	−0.305	0.055
Government expenditure	−0.000	−0.002	0.001
Government expenditure (SEK)	−2.6	−39.0	12.2
Welfare	−0.002	0.001	−0.004
<i>Full-scale roll-out</i>			
Unemployment (measured in %)	−0.197	−0.924	0.058
Government expenditure	−0.001	−0.005	0.001
Government expenditure (SEK)	−14.4	−118.0	22.4
Welfare	−0.006	0.004	−0.012

Note: Outcomes are normalized with the monthly output per worker set to 1.

C.5. Delayed vacancy response

The equilibrium conditions presume that the labor market is in the new steady state during the experiment, which includes the full demand side response with complete adjustment of vacancies. However, firms may not have time to observe and react to the new economic environment in the short run. Therefore we could expect further adjustments of vacancies in the long run if the program was made permanent, which would lead to more positive effects of JSA since additional vacancies would increase the job finding rate and push down the unemployment rate. To explore the implications of this, we estimate an alternative version of our model under the assumption that job-search activity adjusts during the experiment, but that firms do not adjust vacancies during the experiment.³²

As expected, this delayed vacancy approach predicts a larger reduction of unemployment and government expenditures (down by about SEK 118 per worker) than in the main analyses. Also, interestingly, the welfare effects are now reversed with positive welfare effects compared to the negative welfare effects found in the main analyses. In fact, the highest level of welfare is now obtained with a full-scale roll-out of the program, suggesting that the JSA program may also be welfare enhancing.

³²This analysis assumes that the participants react to the program, and that both participants and non-participants realize and respond to the fact that the return to an application is lower during the experiment. However, we assume that firms do not observe that the average wage and matching rates in the labor market have changed, and thus, post vacancies at the same rate as before the experiment (market tightness remains at the pre-experiment level). Specifically, we first estimate the model under the assumption that the labor market is in a steady-state equilibrium before the experiment, and then estimate the model for the experiment keeping market tightness fixed at the pre-experiment level. The number of applications and wages adjust during the experiment and are given by equations (6), (8) and (16). Given the resulting estimates of the model, we then perform simulations allowing for the full demand side response with a complete adjustment of vacancies.

C.6. Sensitivity analyses

In the main analyses, the share of participants is set to the share in the target population. However, if non-participants outside the experiment, such as the stock of unemployed before the experiment and the inflow to the non-target population, are also subject to displacement effects, this implies a lower treatment share. We therefore explore setting the share of participants to 0.20. The displacement effects are set to the same level as for the non-participants in the experiment, which means that we extrapolate the displacement effects for the non-participants in the target population to the non-participants outside the target population. By construction, this leads to less positive effects. Table C-3 shows that the share of participants in the main analyses (31%) implies a non-negligible reduction of the unemployment rate (0.2 percentage points) of a full-scale roll-out of the program. In contrast, a treatment share of 20% implies an increase of the unemployment rate.